

PEARSON



Digital Systems

Principles and Applications

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Chapter 9 Objectives

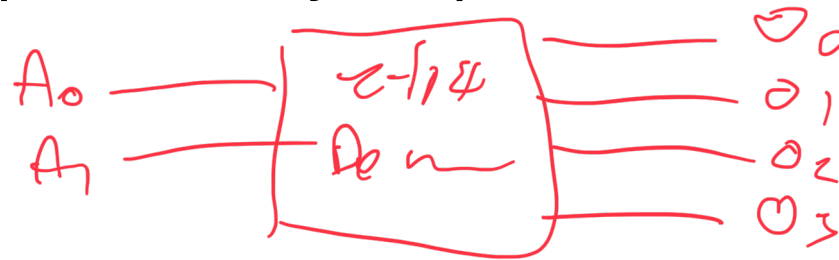
- *Selected areas covered in this chapter:*
 - Analyzing/using decoders & encoders in circuits.
 - Advantages and disadvantages of LEDs and LCDs.
 - Observation/analysis techniques to troubleshoot digital circuits.
 - Operation of multiplexers and demultiplexers in circuit applications.
 - Comparing two binary numbers by using the magnitude comparator circuit.
 - Function and operation of code converters.
 - **Precautions** when connecting digital circuits using the data bus concept.
 - Using HDL to implement MSI logic circuits equivalents.

ข้อควรระวัง

- Digital systems obtain data and information continuously operated on in some manner:
 - *Decoding/encoding.*
 - *Multiplexing/demultiplexing,.*
 - *Comparison; Code conversion; Data busing.*
- These and other operations have been ^{สะดวก} facilitated by the ^{ความพร้อม} availability of numerous ICs in the MSI (medium-scale-integration) category.

9-1 Decoders

- Decoders are used when an output or a group of outputs is to be activated only on the ^{เหตุการณ์} occurrence of a specific combination of input levels.
 - Often provided by outputs of a counter or a register.

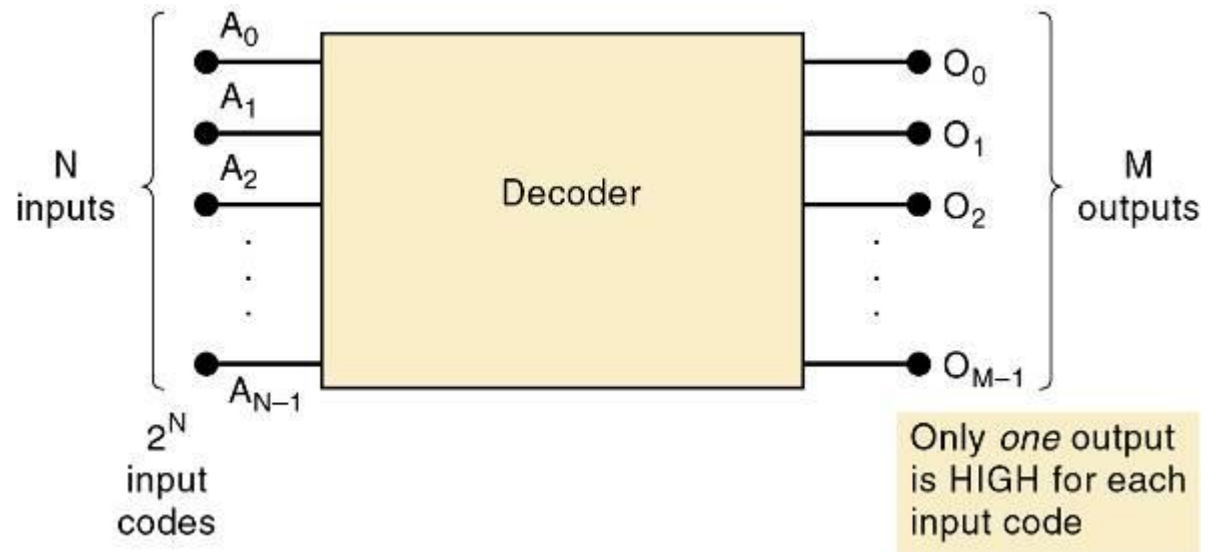


A_1	A_0	O_3	O_2	O_1	O_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

9-1 Decoders

- A **decoder** accepts a set of inputs that represents a binary number—activating only the output that **สอดคล้อง** **corresponds** to the input number.

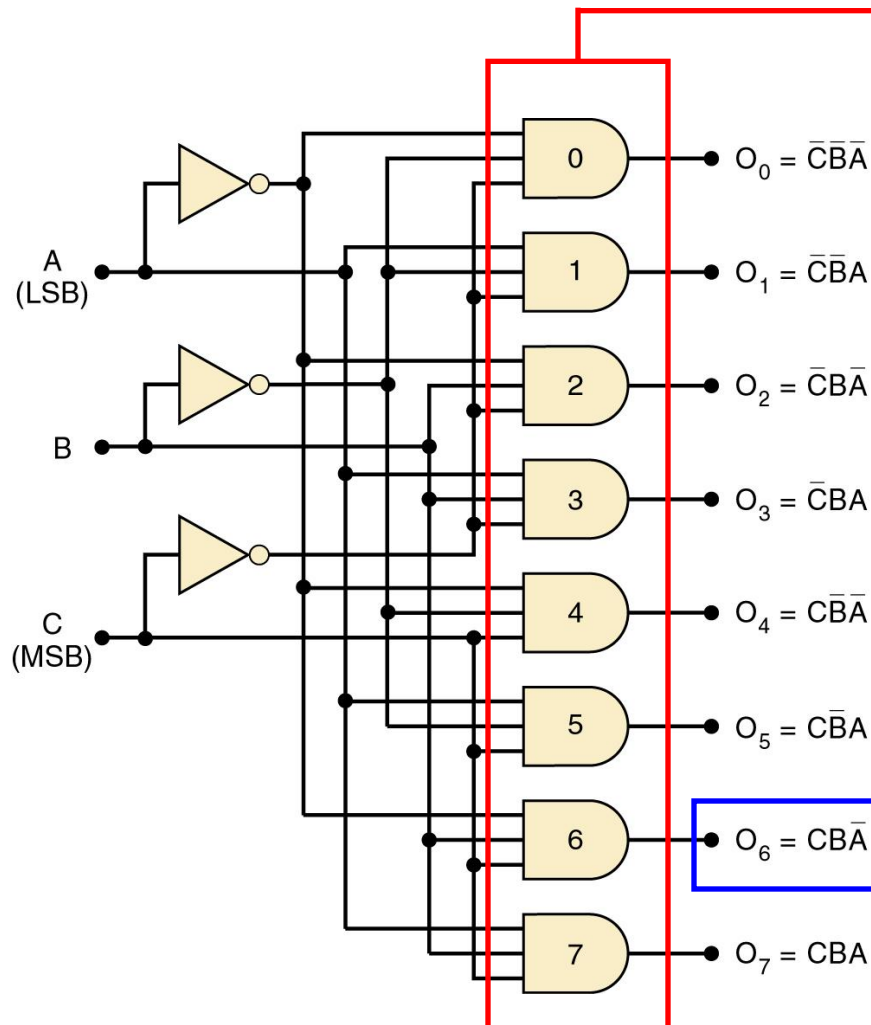
For each of these input combinations, only one of the M outputs will be active (HIGH); all the other outputs are LOW.



Many decoders are designed to produce active-LOW outputs, where only the selected output is LOW while all others are HIGH.

9-1 Decoders

Circuitry for a decoder with three inputs and 8 outputs.



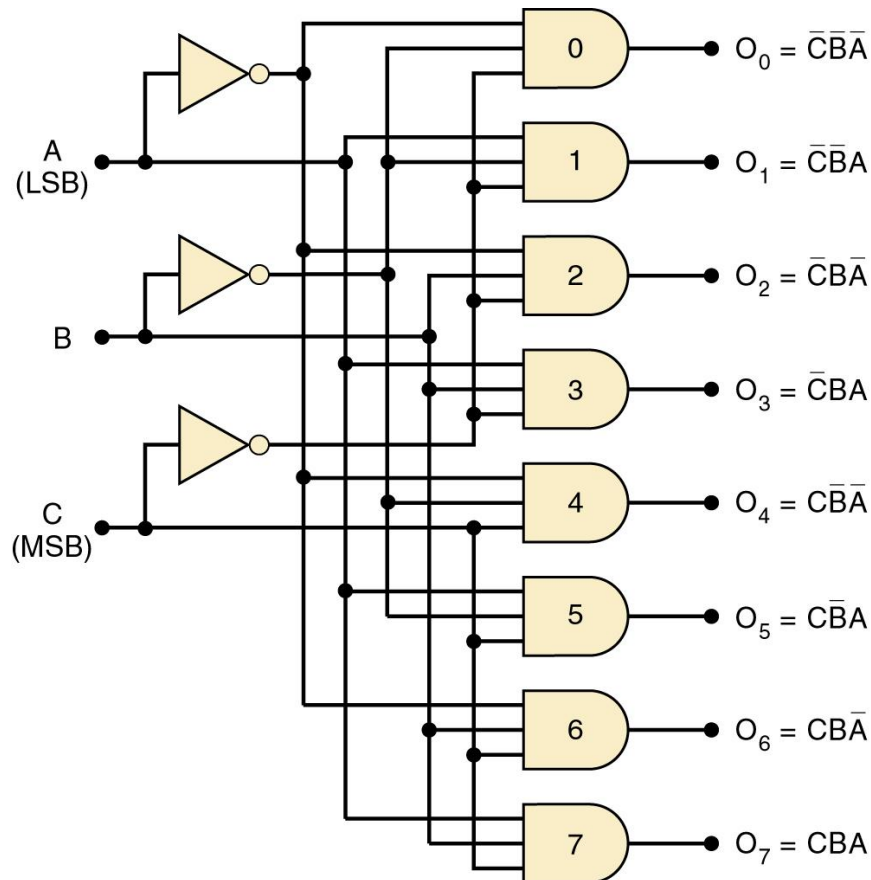
It uses all **AND** gates, so outputs are active-HIGH

C	B	A	O ₇	O ₆	O ₅	O ₄	O ₃	O ₂	O ₁	O ₀
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

Output O₆ goes HIGH only when $CBA\ 110_2 = 6_{10}$.

9-1 Decoders

Circuitry for a decoder with three inputs and 8 outputs.



This can be called a *3-line-to-8-line decoder*—it has three input lines and eight output lines.

Also called a *binary-to-octal decoder* or *converter*—taking three-bit binary input code and activating one of eight (octal) outputs.

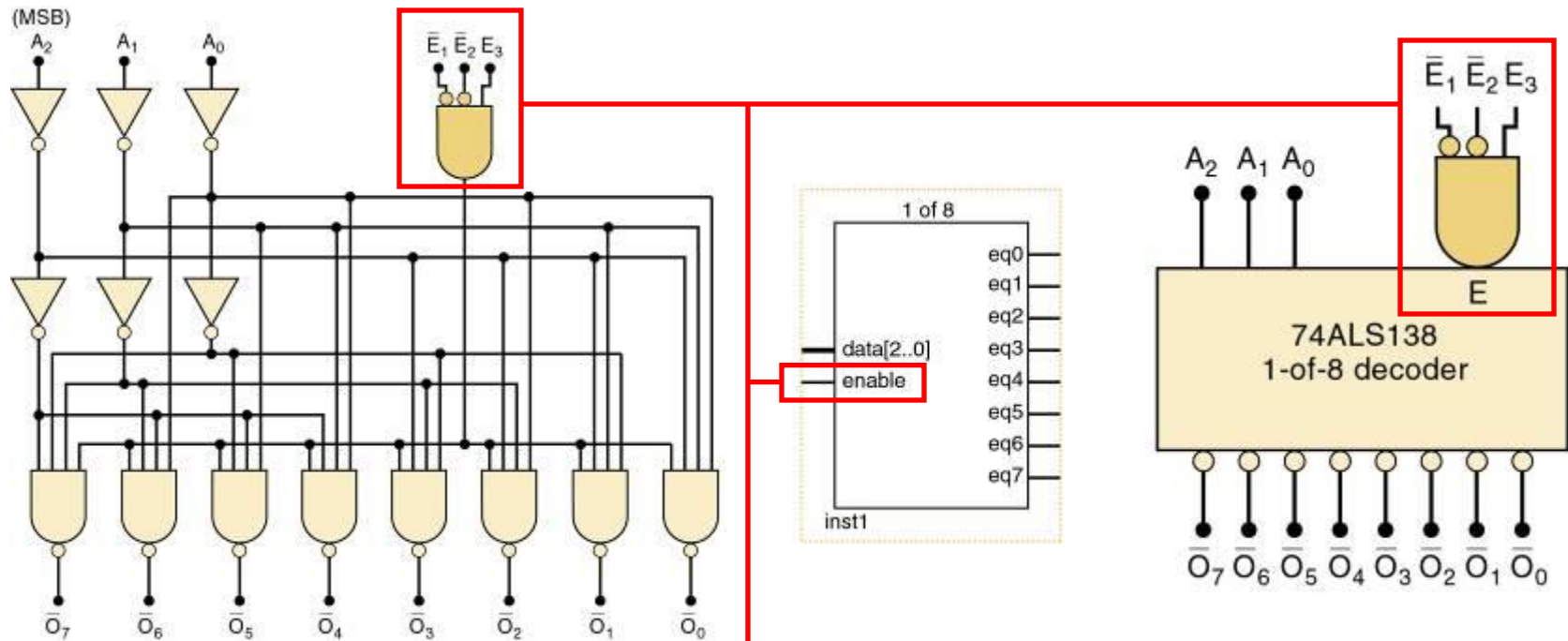
Also referred to as a *1-of-8 decoder*—only 1 of the 8 outputs is activated at one time.

9-1 Decoders

- Some decoders have one or more enable inputs used to control the operation of the decoder.
 - The decoder is enabled only if *ENABLE* is HIGH.
- With common *ENABLE* line connected to a fourth input of each gate:
 - If *ENABLE* is HIGH, the decoder functions normally.
 - A, B, C input will ^{เป็นตัวกำหนด} determine which output is HIGH.
 - If *ENABLE* is LOW, all outputs will be forced LOW.
 - Regardless of the levels at the A, B, C inputs.

9-1 Decoders

The 74ALS138 decoder.



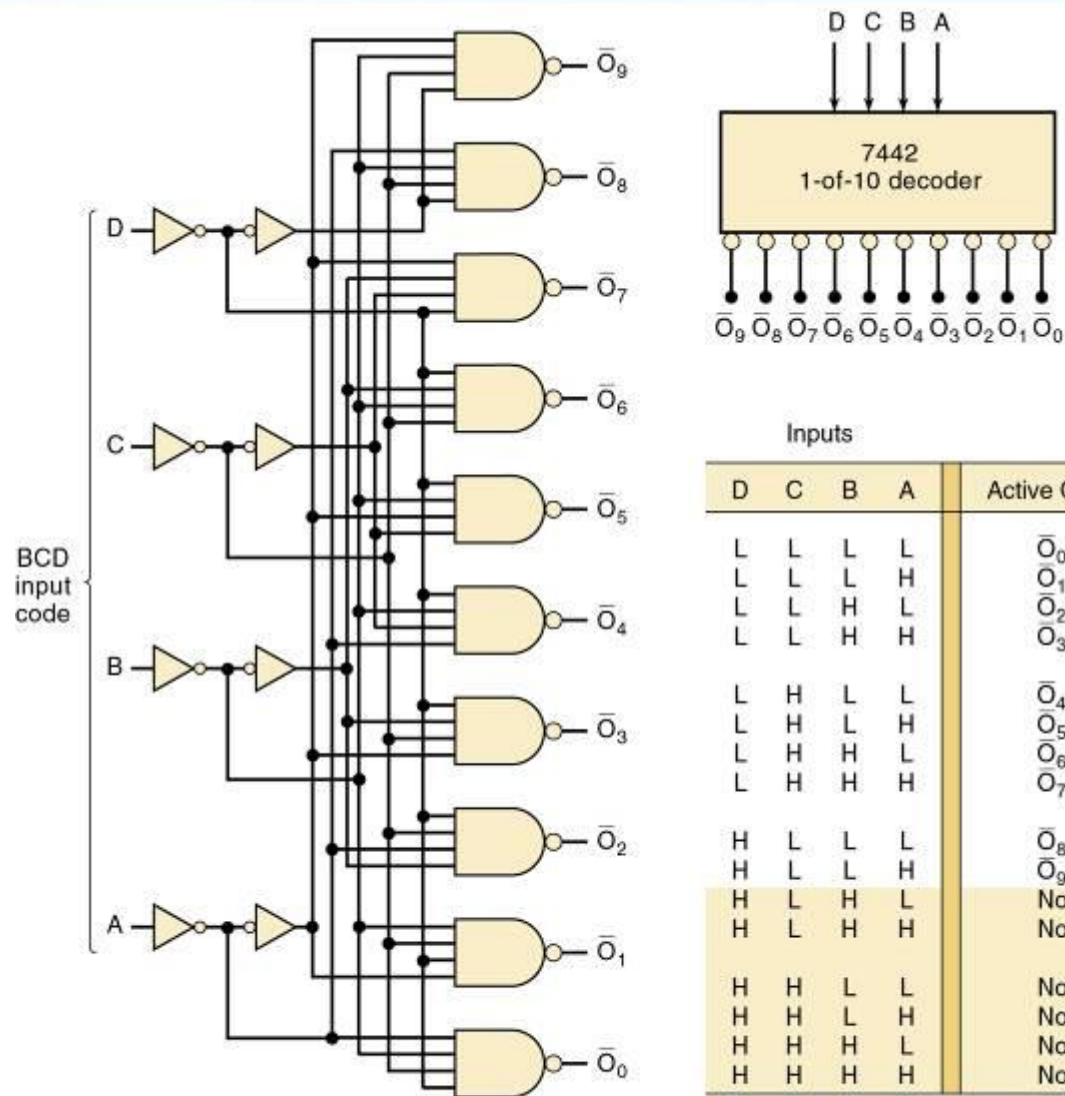
**ENABLE
inputs**

$\bar{E}_1$	$\bar{E}_2$	E_3	Outputs
0	0	1	Respond to input code $A_2A_1A_0$
1	X	X	Disabled – all HIGH
X	1	X	Disabled – all HIGH
X	X	0	Disabled – all HIGH

9-1 Decoders

7442 BCD-to-decimal decoder.

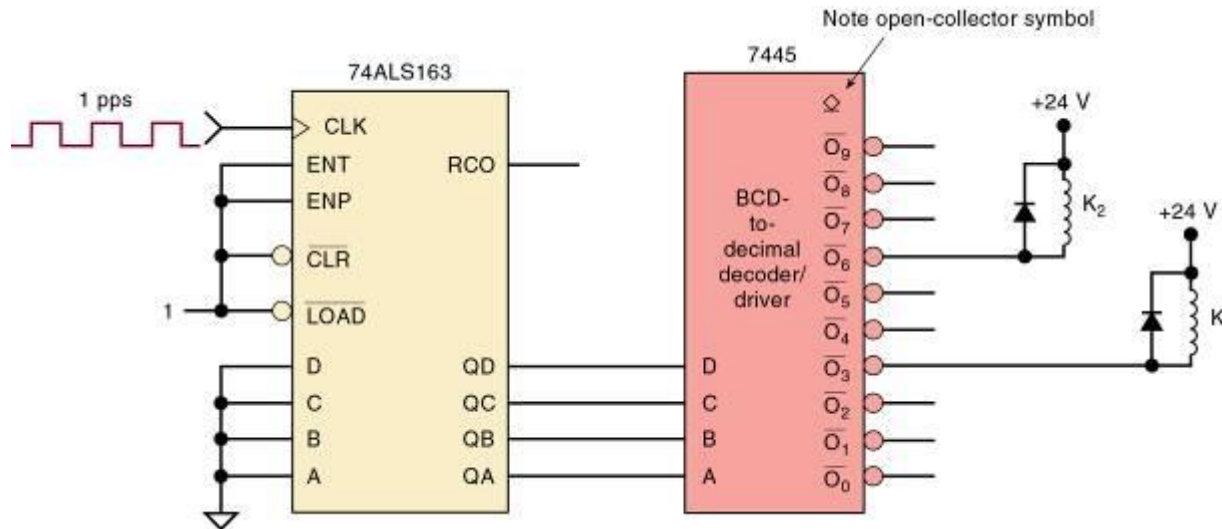
This decoder
does *not* have
an enable input.



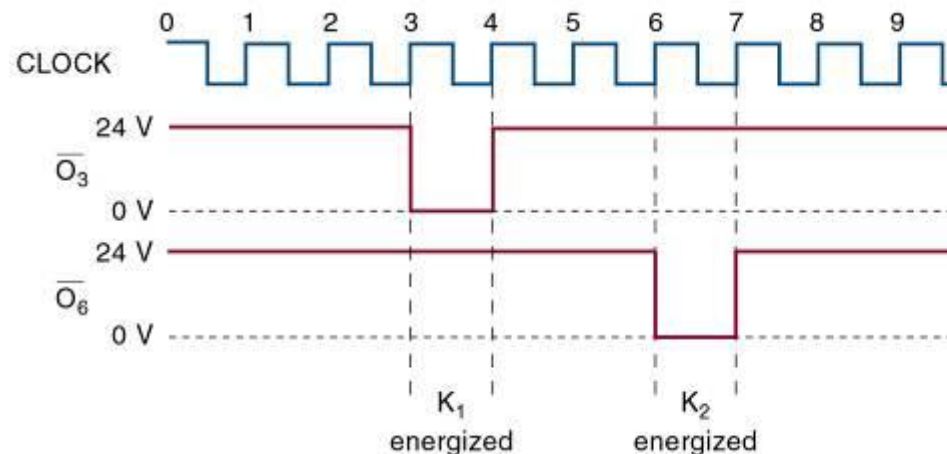
H = HIGH Voltage Level
L = LOW Voltage Level

9-1 Decoders

7445 BCD-to-decimal decoder/driver.



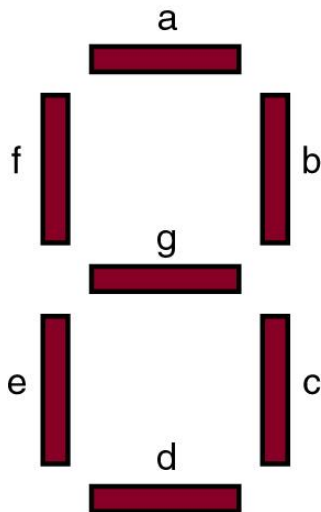
Suitable for directly driving loads such as indicator LEDs or lamps, relays, or dc motors.



Termed a *driver* because this IC has open-collector outputs that operate at higher current/voltage limits than a normal TTL output.

9-2 BCD-to-7 Segment Decoder/Drivers

- The 7-segment display is a common way to display decimal or hexadecimal characters.
 - การจัดเรียงแบบหนึ่งที่พบบ่อย
One common arrangement uses light emitting diodes (LEDs) for each segment.



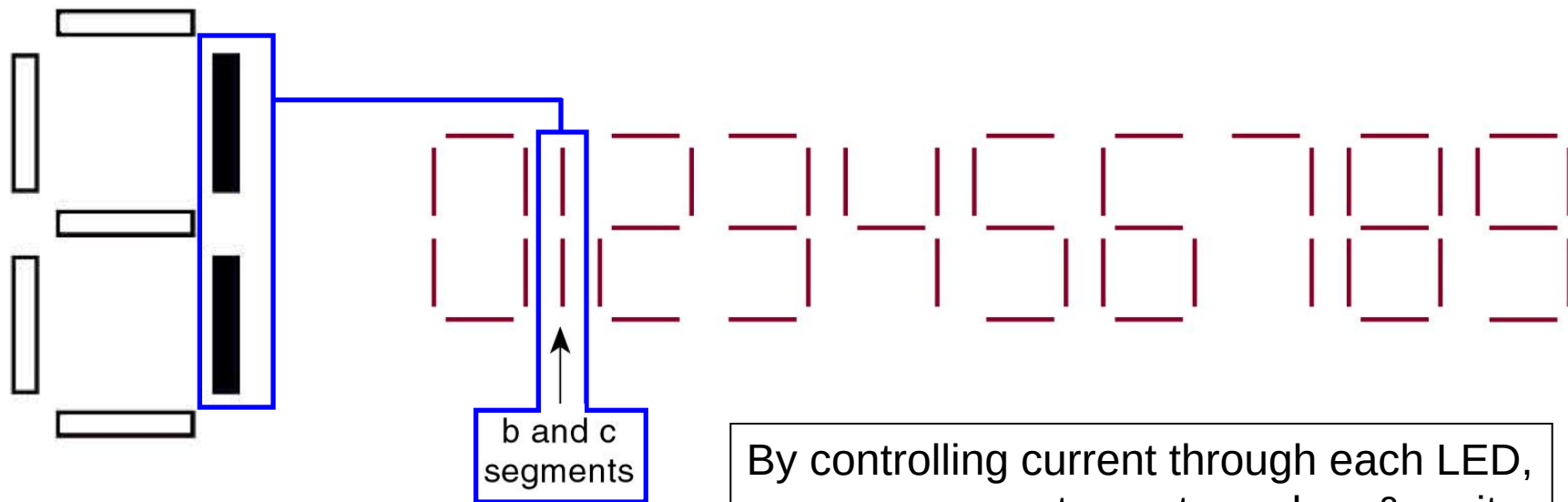
Diodes allow current to flow in one direction, but block flow in the other direction.



When the LED anode is more positive than the cathode by approximately 2 V, the LED will light up.

9-2 BCD-to-7 Segment Decoder/Drivers

- The 7-segment display is a common way to display decimal or hexadecimal characters.
 - One common arrangement uses light emitting diodes (LEDs) for each segment.

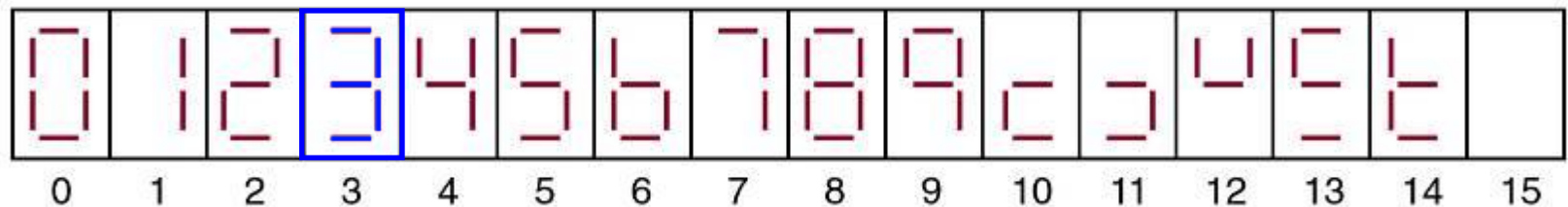
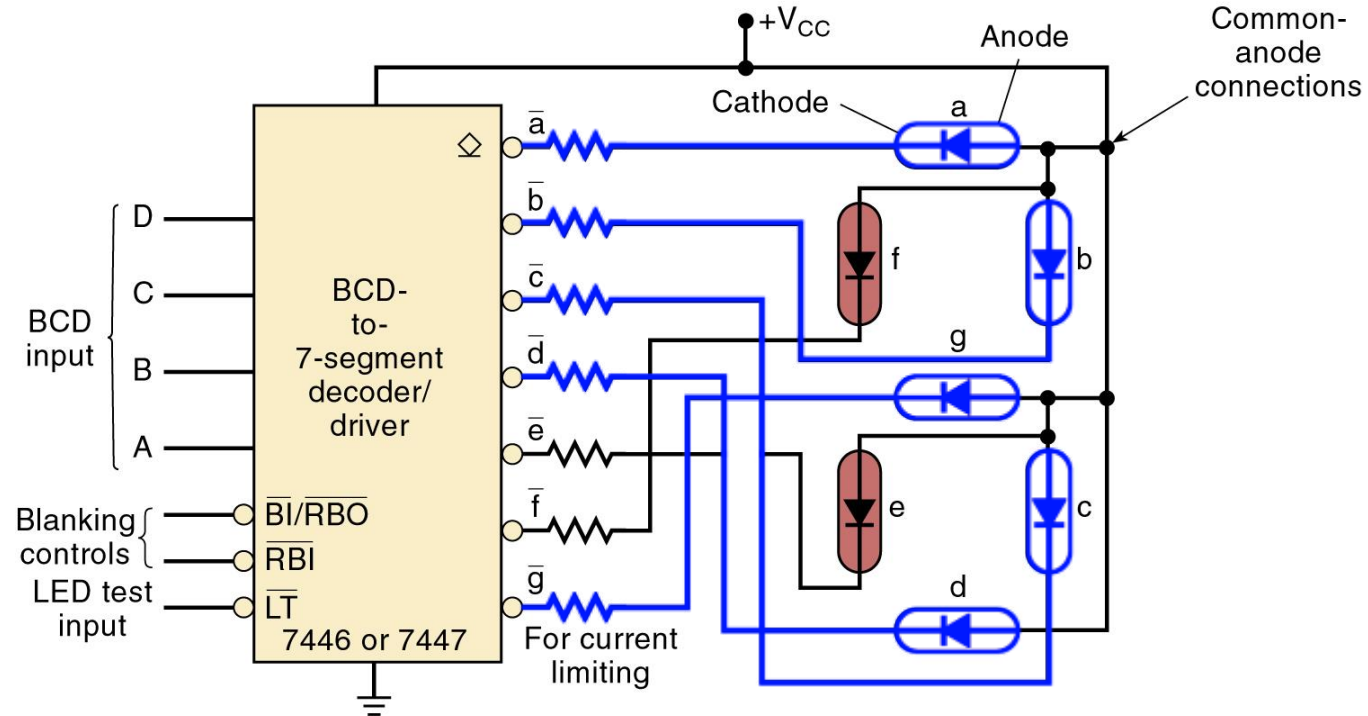


By controlling current through each LED, some segments are turned on & emit light, while others are turned off, which generates the desired character pattern.

9-2 BCD-to-7 Segment Decoder/Drivers

BCD-to-7-segment decoder/driver.

The 7446/47 activates specific segment patterns in response to input codes

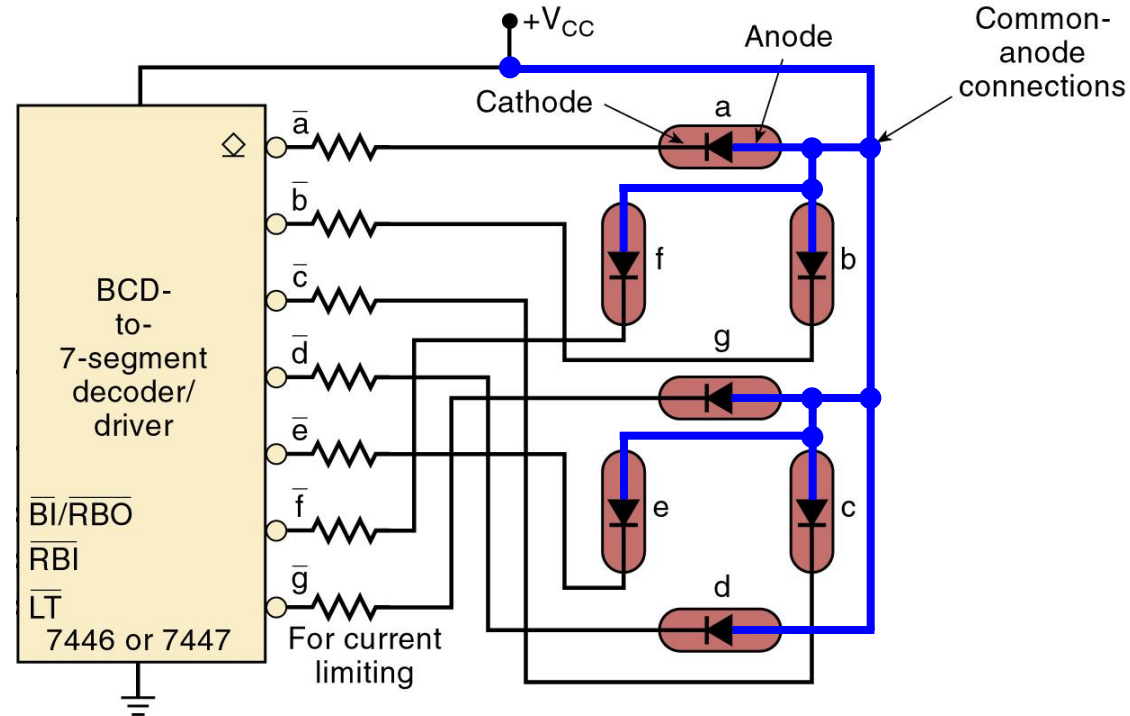


9-2 BCD-to-7 Segment Decoder/Drivers

BCD-to-7-segment decoder/driver.

This is a **common-anode** LED display.

The anodes all of segments are tied together to V_{CC} .



Another type uses a **common-cathode** method, with each segment requiring 10 to 20 mA of current. TTL/CMOS devices are normally not used to drive a common-cathode display directly—a transistor interface circuit is often used

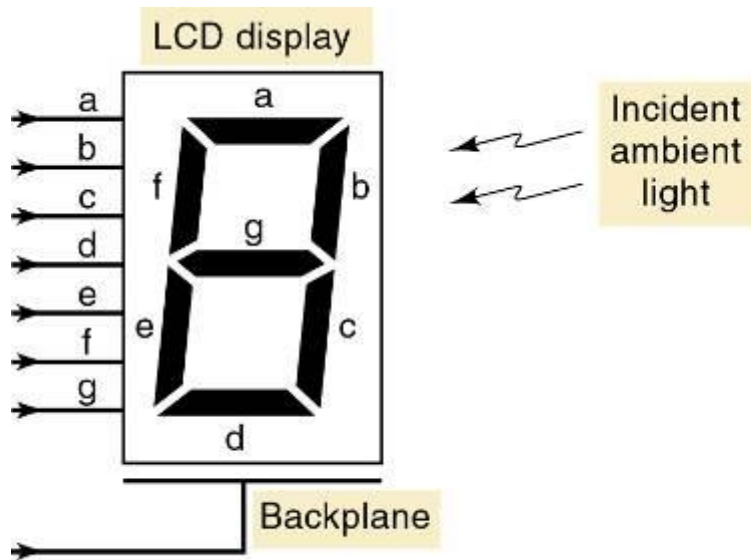
9-3 Liquid Crystal Displays

- A liquid-crystal display (**LCD**) controls reflection of available light.
 - The light may be ambient (room, sun) light.
 - *Reflective* LCDs use ambient light.
 - Available light might be provided by a small light source—part of the display unit.
 - *Backlit* LCDs use this method.
- LCDs have gained wide acceptance due to their very low power consumption—compared to LEDs.
 - LEDs have the advantage of a much brighter display that is easily visible in dark or poorly lit areas.

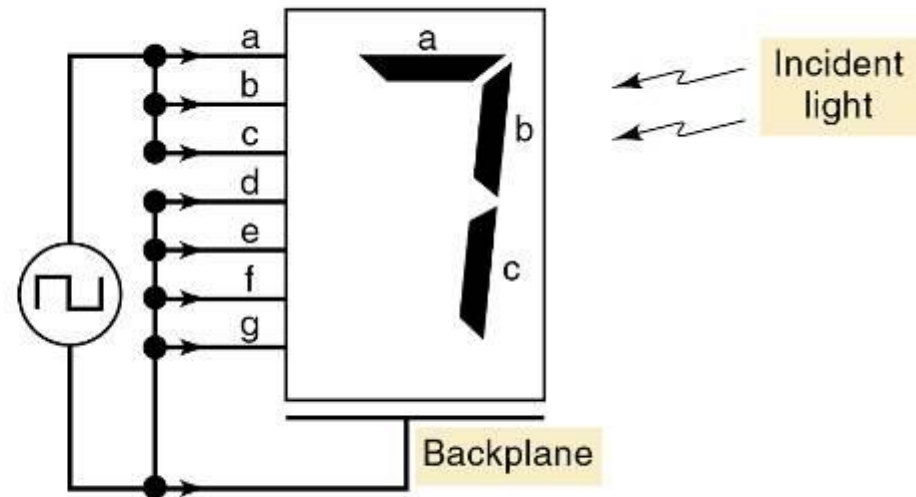
9-3 Liquid Crystal Displays

Liquid-crystal display.

Basic arrangement.



Applying voltage between the segment and the backplane turns the segment ON.

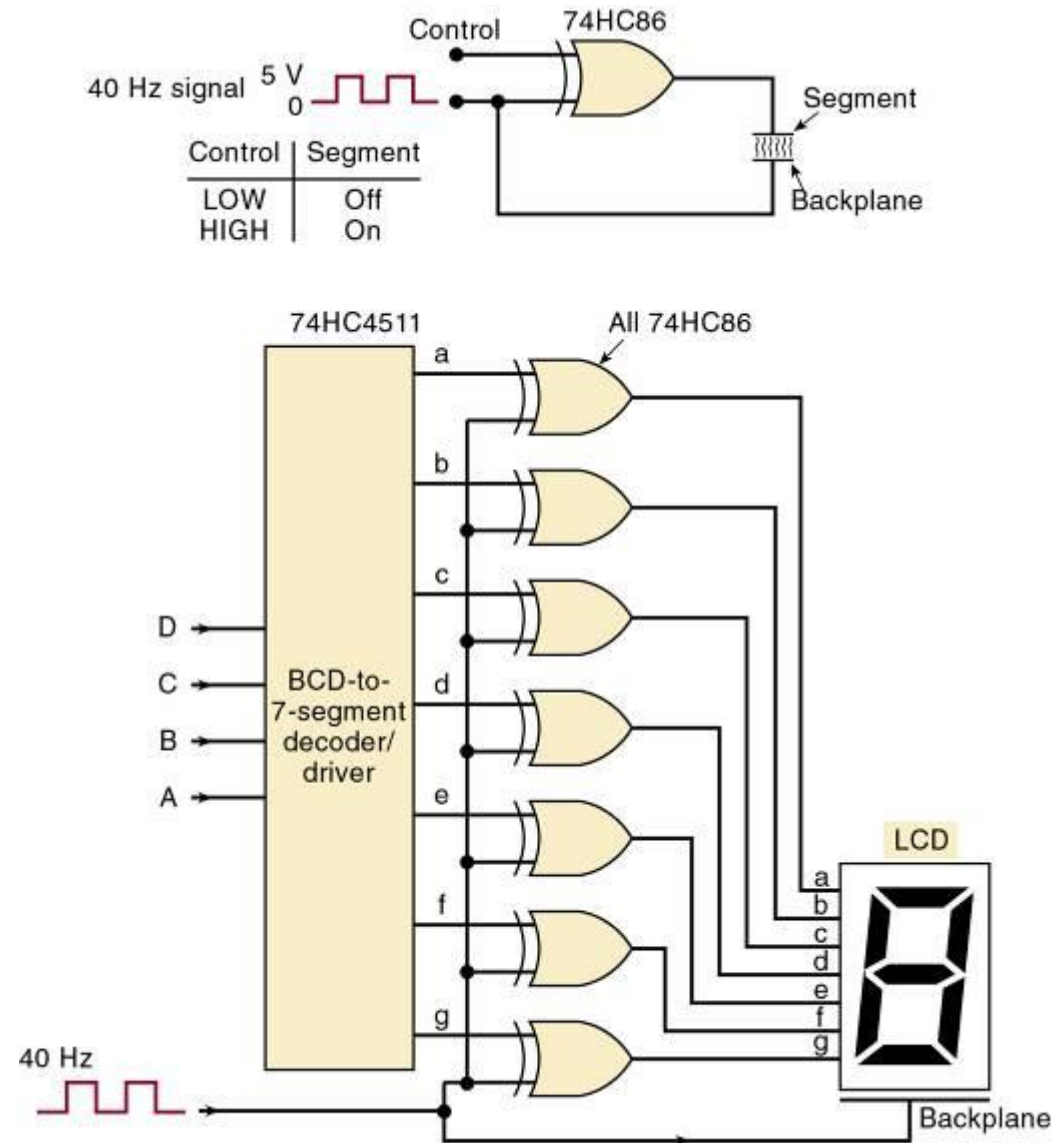


Zero voltage turns the segment OFF.

9-3 Liquid Crystal Displays

Driving a 7-segment display.

It is common to produce required ac voltage by applying out-of-phase square waves to the segment and the backplane.



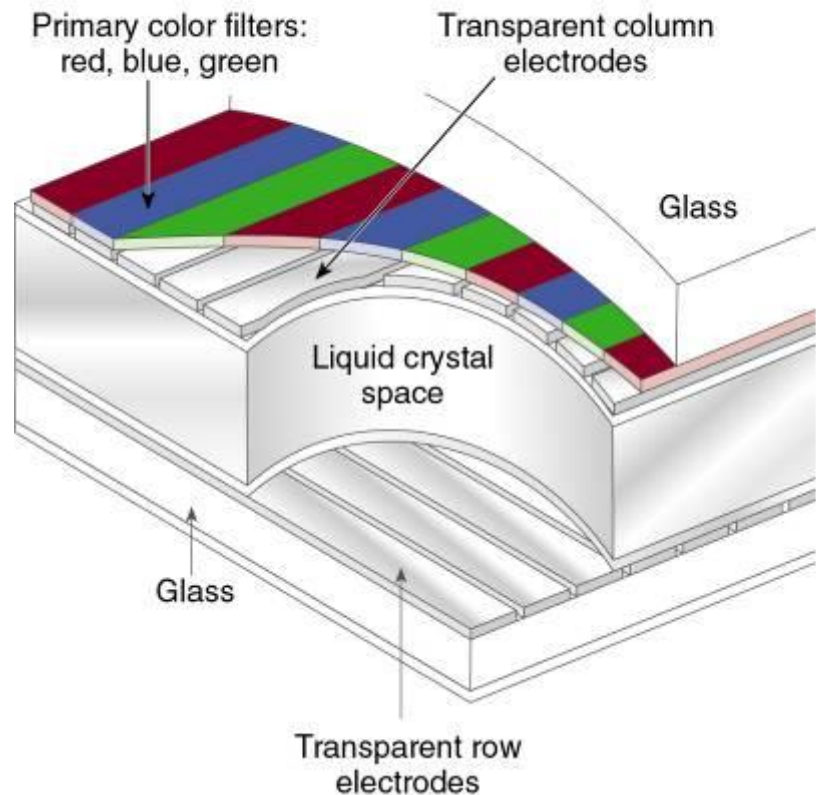
9-3 Liquid Crystal Displays

- Liquid crystals are available as multidigit 7-segment decimal numeric displays.
 - Alphanumeric LCD modules are available in many formats—up to 4 lines by 40 characters.
- Other LCD modules allow the user to create a graphical display by controlling individual dots on the screen called **pixels**.
- Larger LCD panels can be scanned at a high rate, producing high quality video motion pictures.
 - Control lines are arranged in rows & columns.
- A digital system must activate a matrix row/column to control the amount of light at each pixel.

9-3 Liquid Crystal Displays

- Each pixel on a color display is made up of three subpixels, which control light passing through a **red**, **green**, or **blue** filter to produce pixel color.

A 640 x 480 LCD screen would have 640 x 3 connections for columns and 480 connections for rows—a total of 2400 connections to the LCD.



9-3 Liquid Crystal Displays

- Older screens are called Twisted Nematic (TN) or Super Twisted Nematic (STN).
 - Referred to as passive LCDs.
- Newer displays are called active matrix TFT LCDs.
 - An active element on the display is used to switch pixels on and off.
- Other display technologies—vacuum fluorescent, gas discharge plasma, and electroluminescence.
 - Optical physics for each of these displays varies.
 - The means of controlling all of them is the same.

9-4 Encoders

- Most decoders accept an input code & produce a HIGH (or LOW) at *one* and *only one* output line.
 - A decoder identifies, recognizes, or detects a particular code.

9-4 Encoders

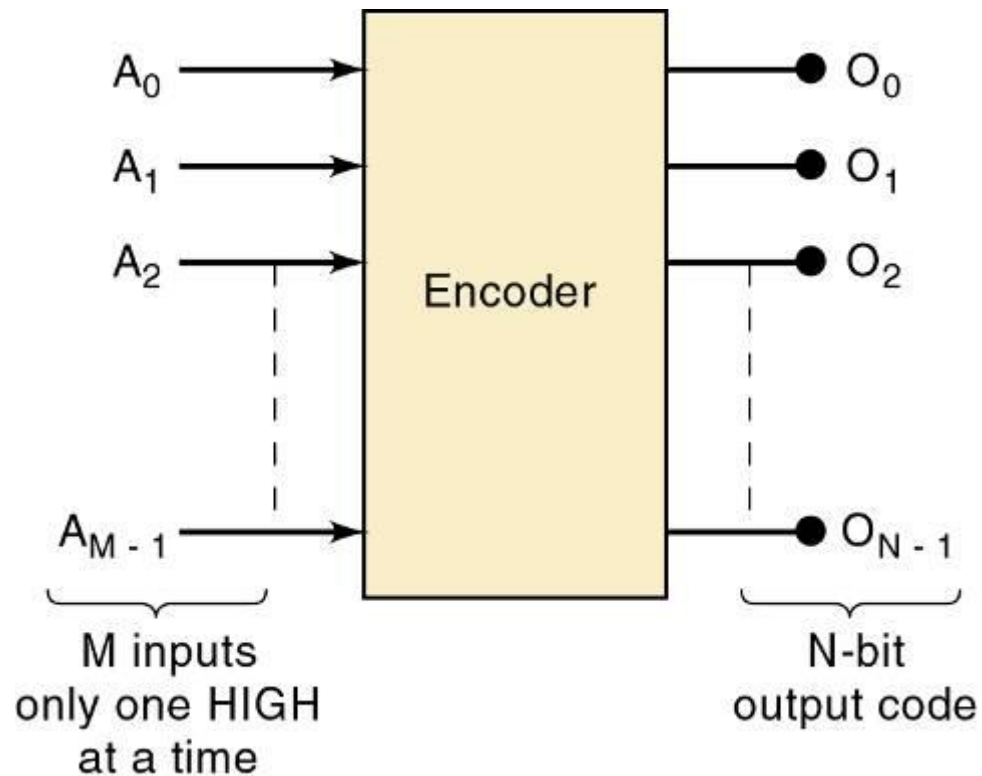
- The opposite of decoding process is **encoding**.
 - Performed by a logic circuit called an **encoder**.

An encoder has a number of input lines, only **one** of which is activated at a given time.

Shown is an encoder with M inputs and N outputs.

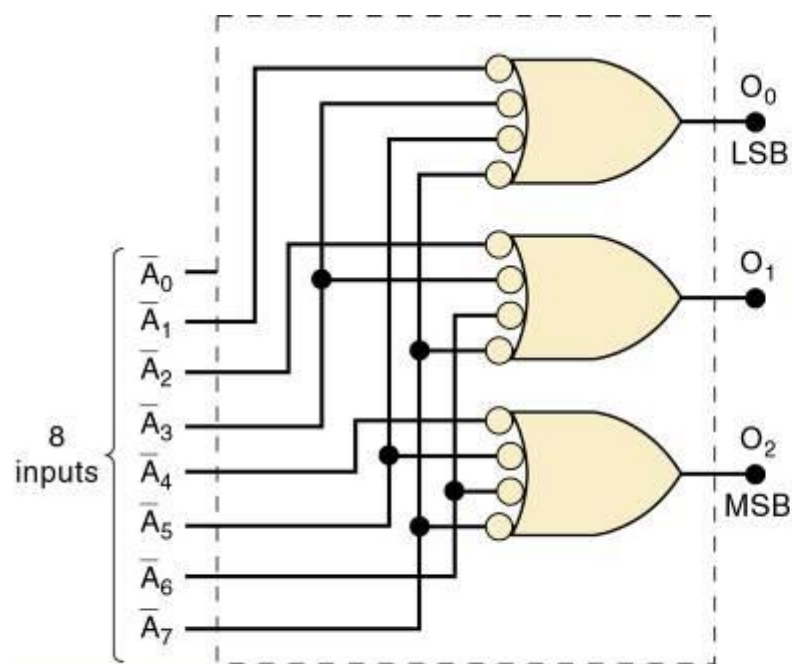
Inputs are active-HIGH, which means that they are normally LOW.

It produces an N -bit output code, depending on which input is activated.



9-4 Encoders

- An *octal-to-binary encoder (8-line-to-3-line encoder)* accepts eight input lines, producing a three-bit output code corresponding to the input.



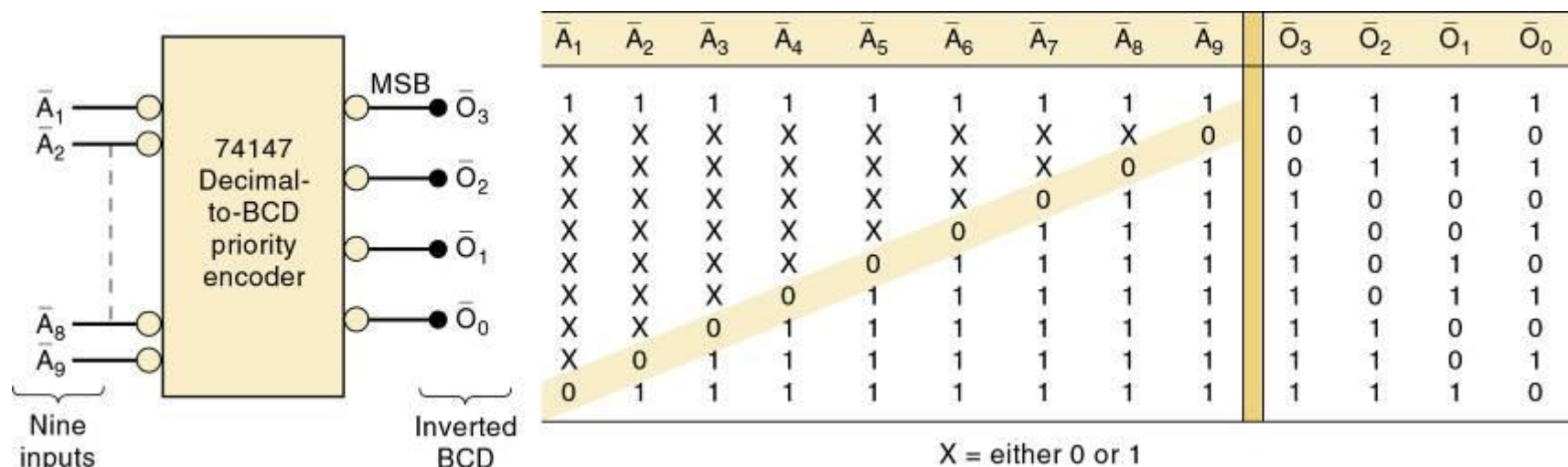
Inputs								Outputs		
$\bar{A}_0$	$\bar{A}_1$	$\bar{A}_2$	$\bar{A}_3$	$\bar{A}_4$	$\bar{A}_5$	$\bar{A}_6$	$\bar{A}_7$	O_2	O_1	O_0
X	1	1	1	1	1	1	1	0	0	0
X	0	1	1	1	1	1	1	0	0	1
X	1	0	1	1	1	1	1	0	1	0
X	1	1	0	1	1	1	1	0	1	1
X	1	1	1	0	1	1	1	1	0	0
X	1	1	1	1	0	1	1	1	0	1
X	1	1	1	1	1	0	1	1	1	0
X	1	1	1	1	1	1	0	1	1	1

*Only one
LOW input
at a time

Logic circuit for an octal-to-binary (8-line-to-3-line) encoder.
Only one input should be active at one time.

9-4 Encoders

- A **priority encoder** ensures that when two or more inputs are activated, the output code will correspond to the highest-numbered input.



It has nine active-LOW inputs represent decimal digits 1 through 9, producing *inverted* BCD code corresponding to the highest-numbered activated input.

9-4 Encoders

- A **switch encoder** can be used when BCD data must be entered manually into a digital system.
 - The 10 switches might be the keyboard switches on a calculator—representing digits 0 through 9.

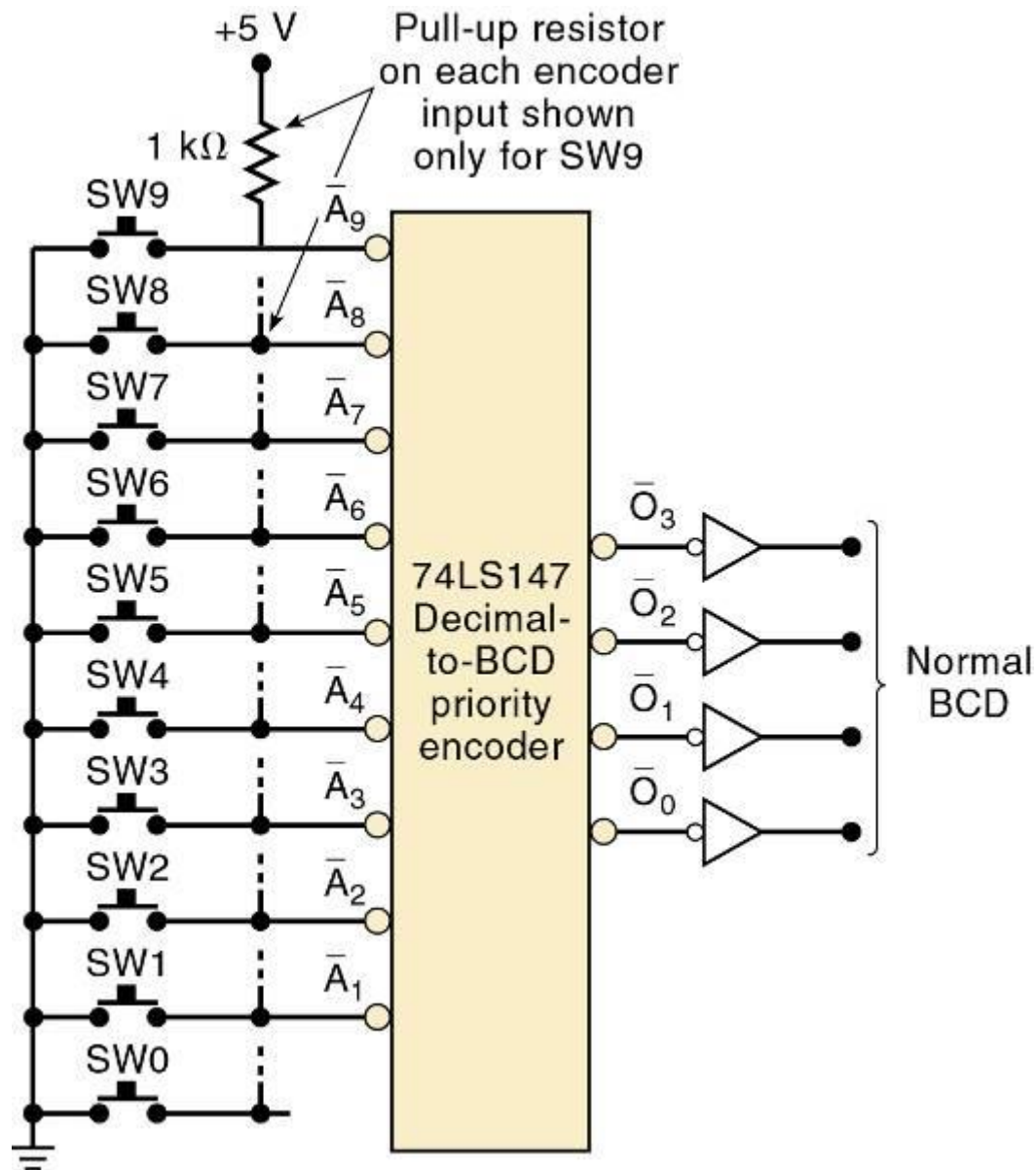
9-4 Encoders

The switches are of the normally open type, so the encoder inputs are all normally HIGH.

BCD output is 0000.

When a key is depressed, the circuit will produce the BCD code for that digit.

The 74LS147 is a *priority* encoder, so simultaneous key depressions produce the BCD code for the *higher-numbered* key.

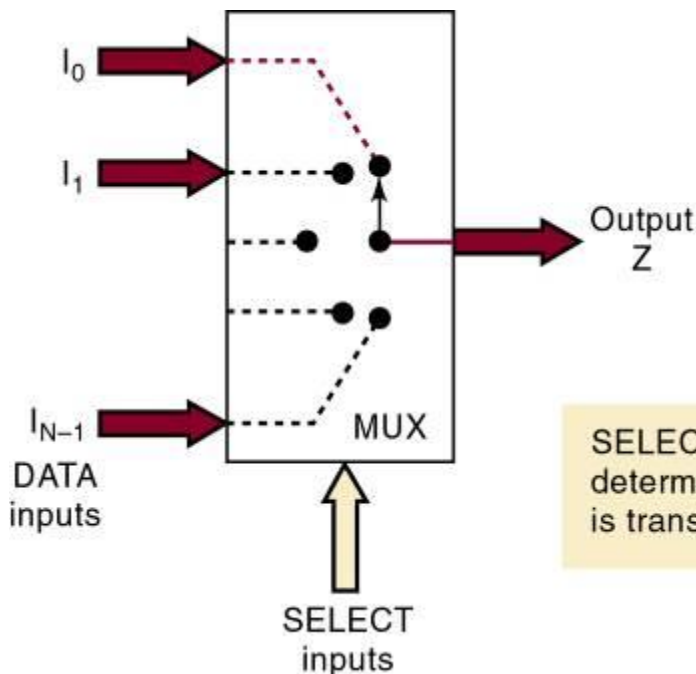


9-5 Troubleshooting

- More complex circuitry increases possible reasons for failure
 - Applying observation and analysis will narrow the focus and simplify testing
- After using observation and analysis to determine the possible faults, repeatedly use the divide and conquer technique to reduce possible causes by half.

9-6 Multiplexers (Data Selectors)

- A **multiplexer (MUX)** selects 1 of N input data sources and transmits the selected data to a single output—called **multiplexing**.
 - A *digital multiplexer* or *data selector* is a logic circuit that performs the same task.

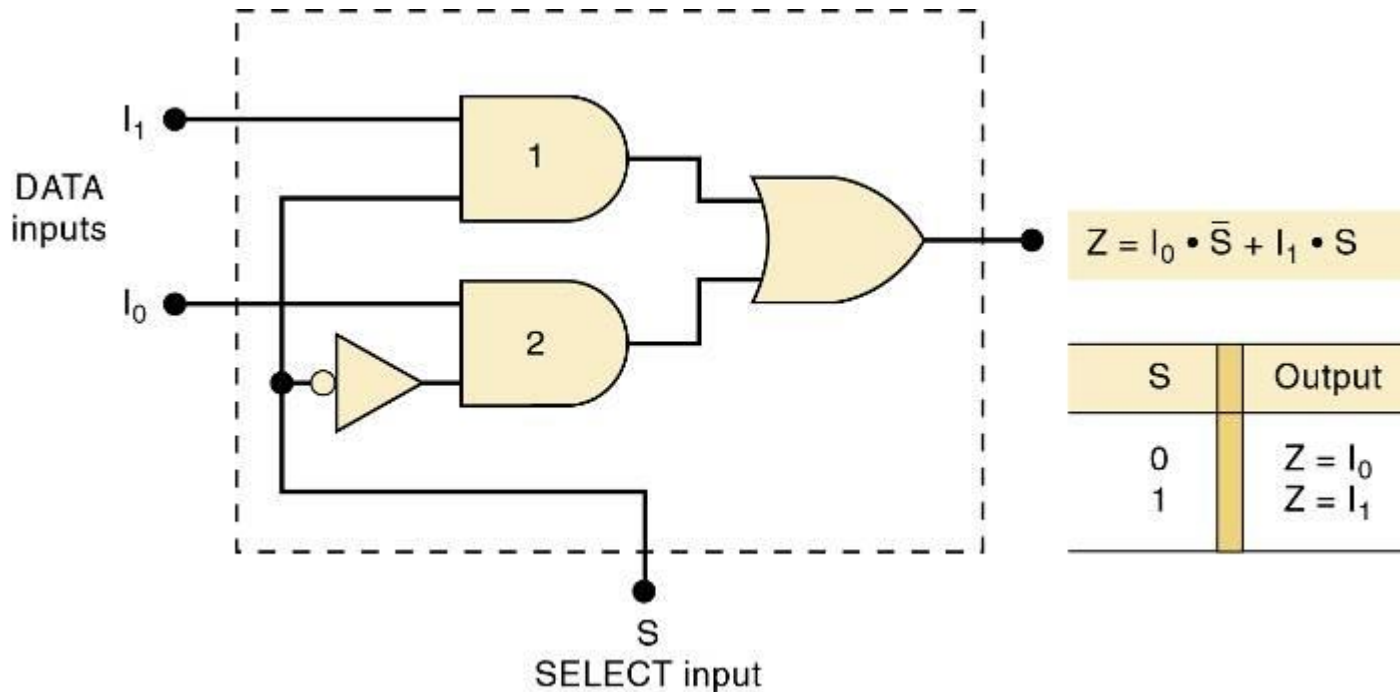


Routing control of desired data input to output by SELECT inputs—referred to as ADDRESS inputs.

SELECT input code determines which input is transmitted to output Z.

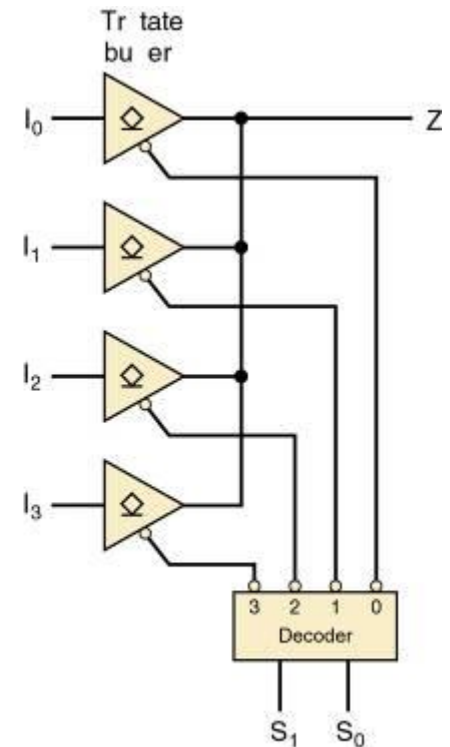
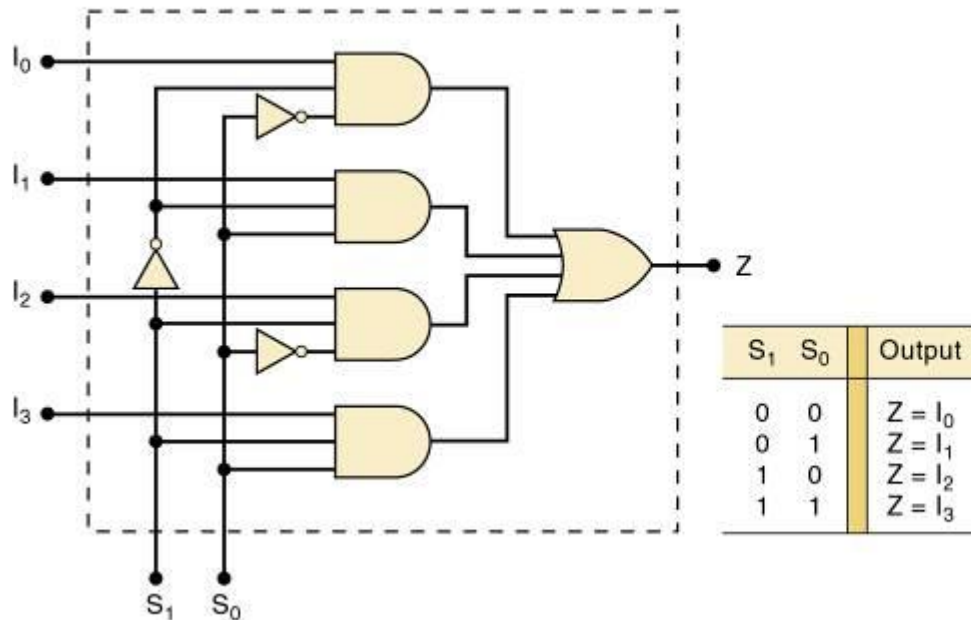
9-6 Multiplexers (Data Selectors)

- A two-input MUX could be used in a digital system that uses two different MASTER CLOCK signals.
 - A high-speed clock in one mode and a slow-speed clock for the other.



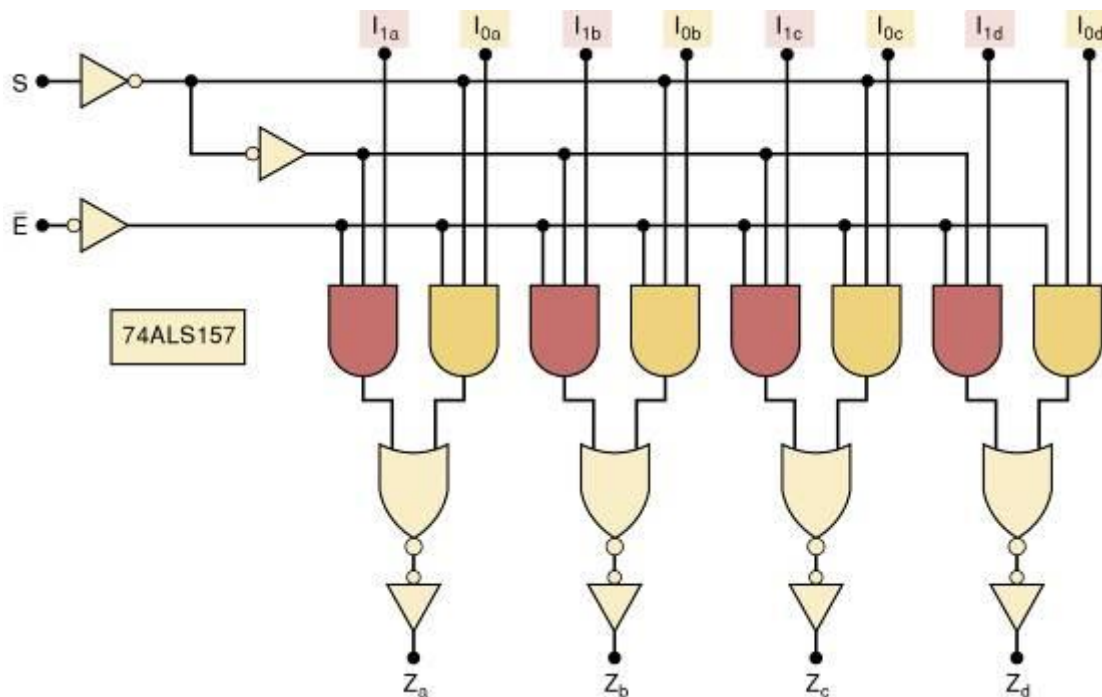
9-6 Multiplexers (Data Selectors)

- Two-, four-, eight-, and 16-input multiplexers are available in the TTL and CMOS logic families.
 - These basic ICs can be combined for multiplexing a larger number of inputs.

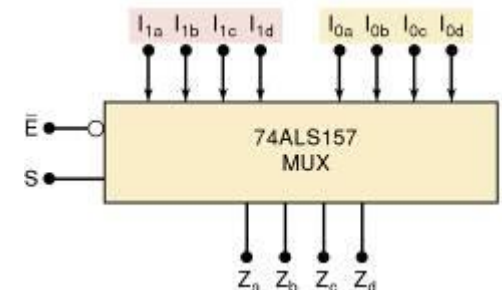
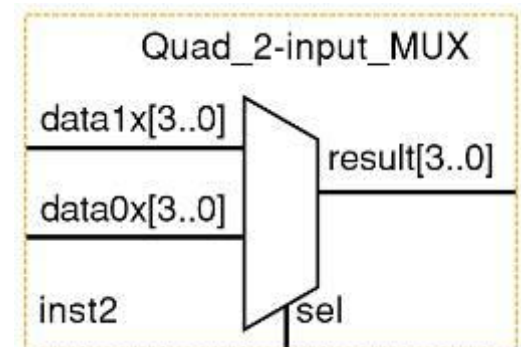


9-6 Multiplexers (Data Selectors)

The 74ALS157 contains four two-input multiplexers



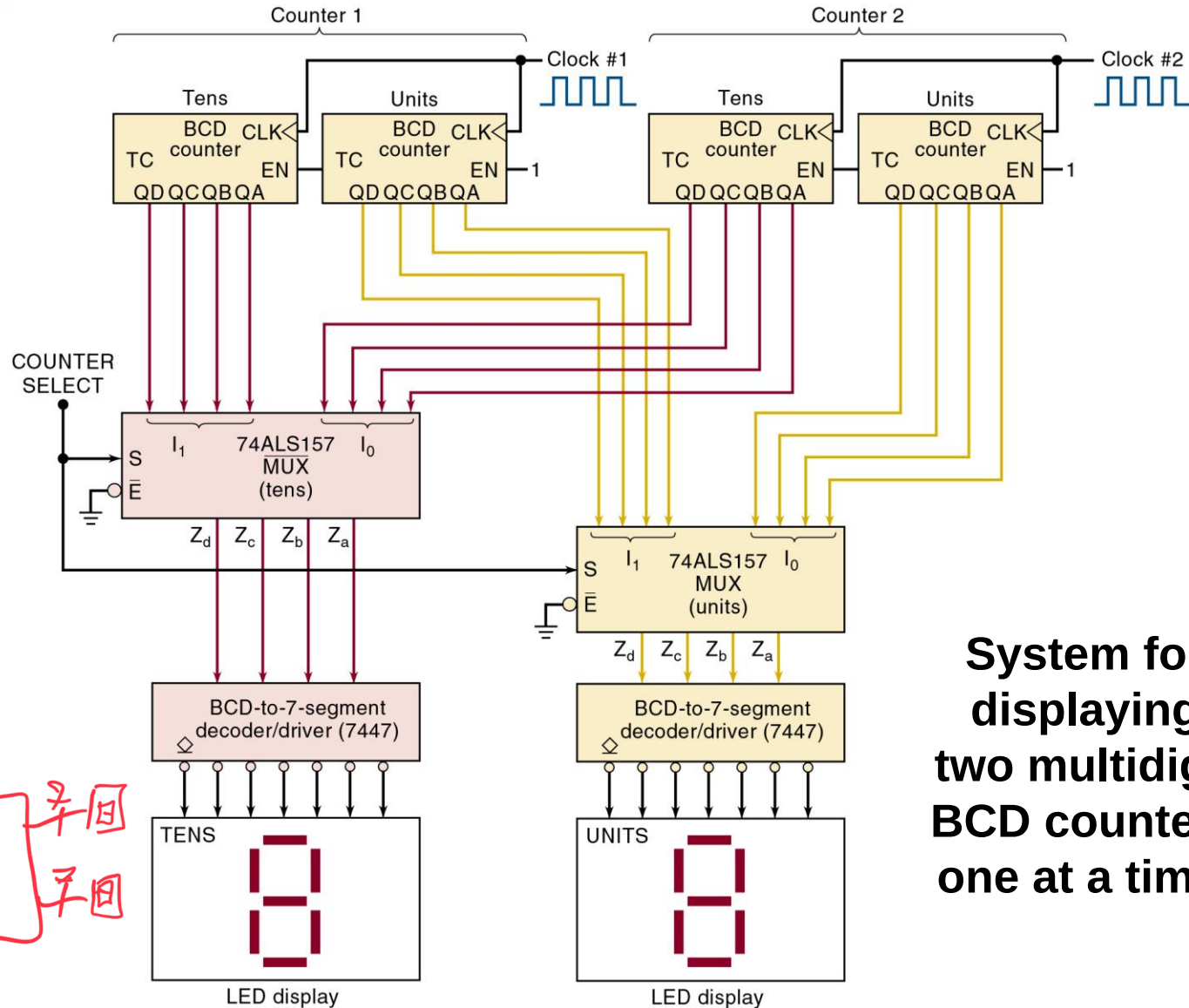
$\bar{E}$	S	Z_a	Z_b	Z_c	Z_d
H	X	L	L	L	L
L	L	I_{0a}	I_{0b}	I_{0c}	I_{0d}
L	H	I_{1a}	I_{1b}	I_{1c}	I_{1d}



9-7 Multiplexer Applications

- Multiplexer circuits find numerous and varied applications in digital systems of all types.
 - Data selection/routing, parallel-to-serial conversion.
 - Operation sequencing.
 - Waveform/logic-function generation.

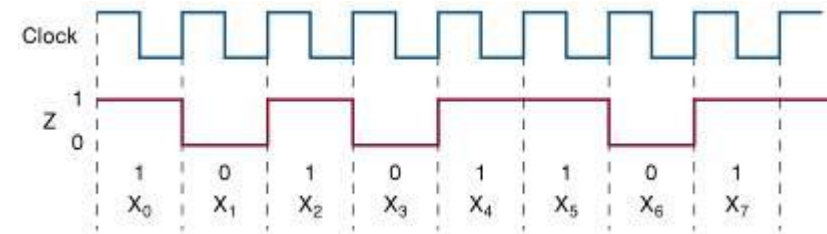
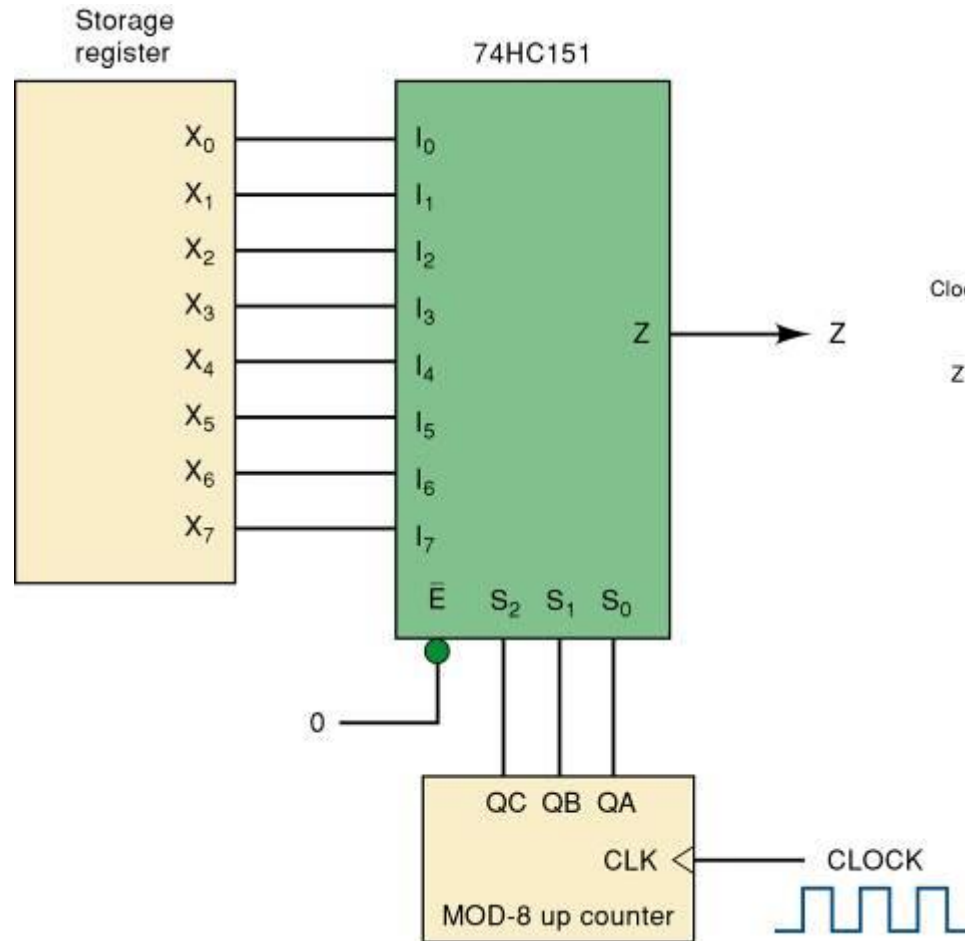
9-7 Multiplexer Applications



System for displaying two multidigit BCD counters one at a time.

9-7 Multiplexer Applications

Parallel-to-serial converter.

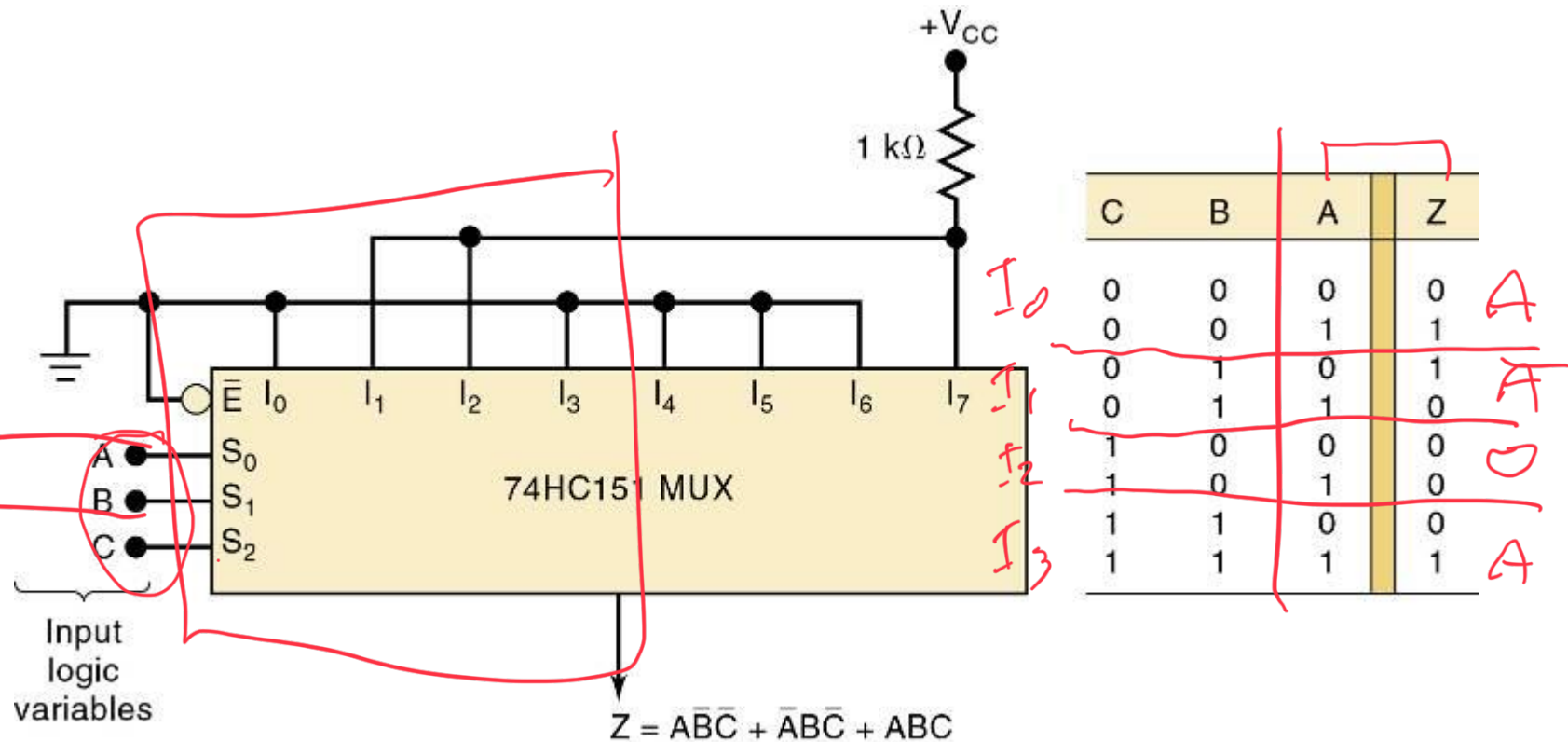


Waveforms for
 $X_7X_6X_5X_4X_3X_2X_1X_0$

10110101

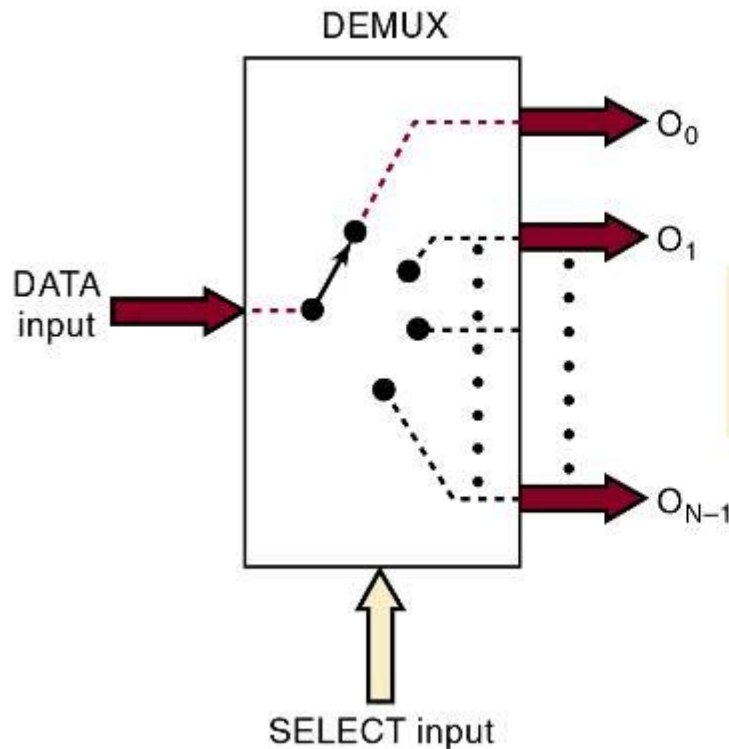
9-7 Multiplexer Applications

Multiplexer used to implement a logic function described by the truth table.



9-8 Demultiplexers (Data Distributors)

- A **demultiplexer (DEMUX)** takes a single input and distributes it over several outputs.
 - The select input code determines to which output the DATA input will be transmitted.

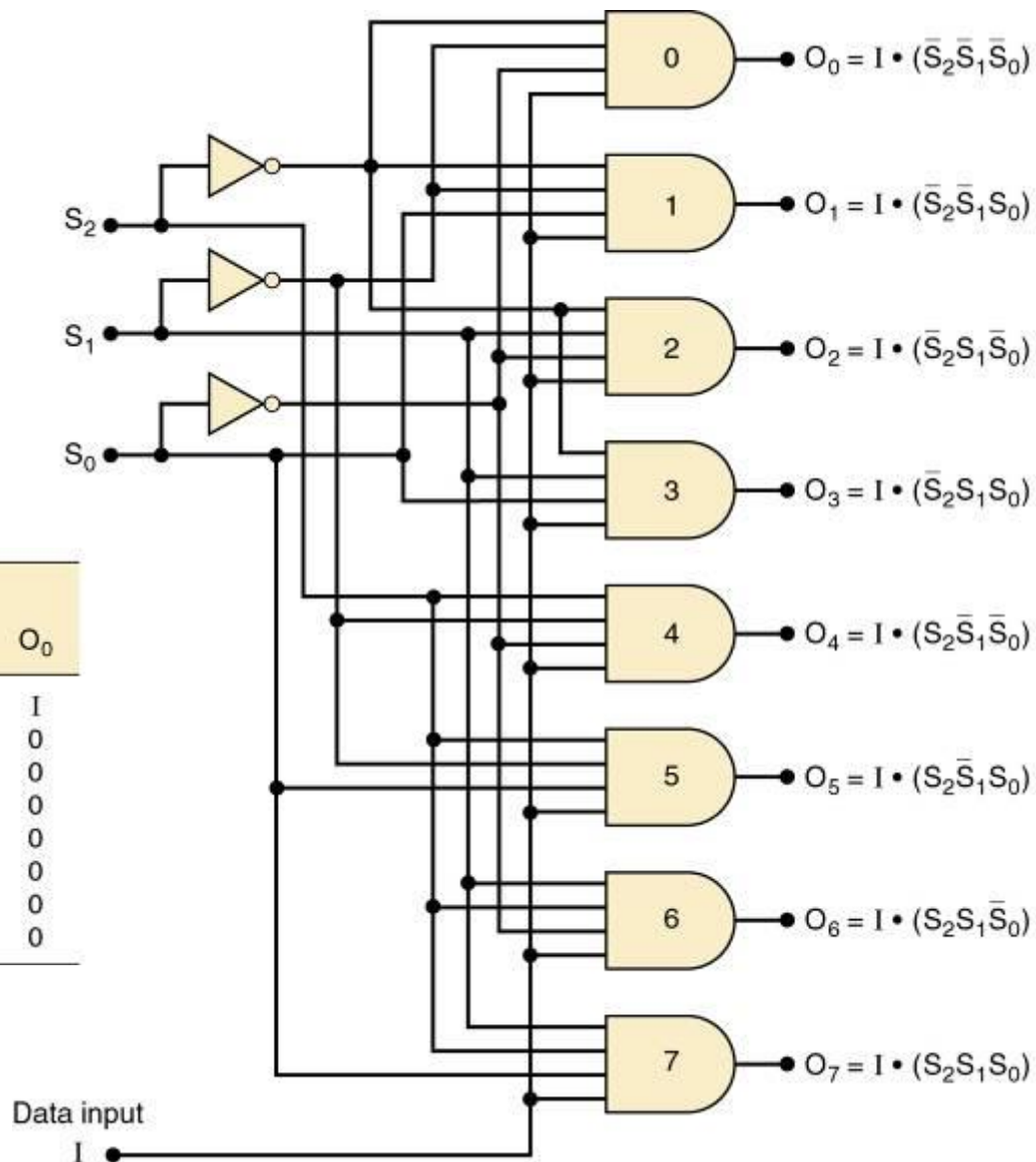


9-8 Demultiplexers (Data Distributors)

A 1 line to 8 line demultiplexer.

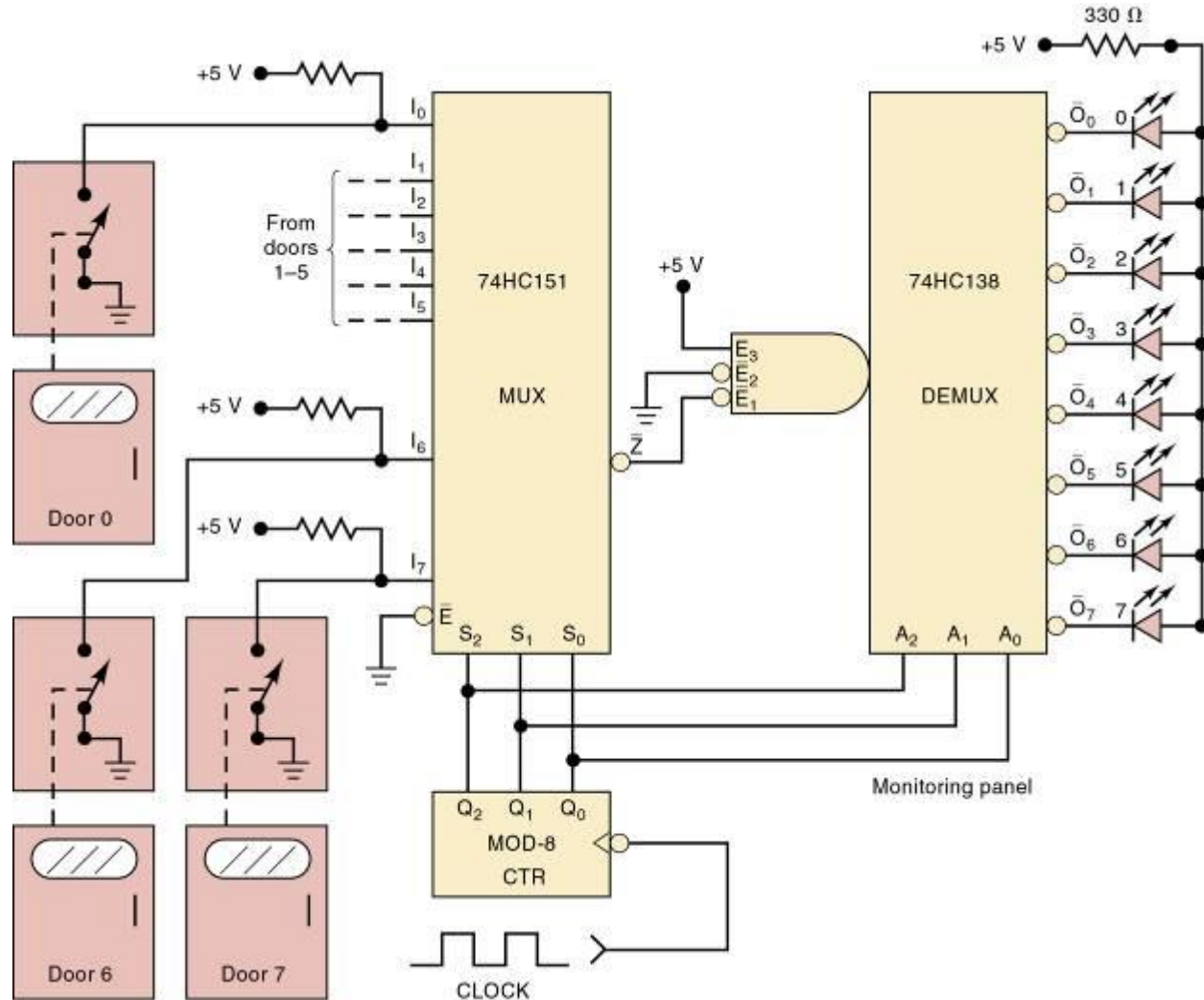
Select Code			Outputs							
S ₂	S ₁	S ₀	O ₇	O ₆	O ₅	O ₄	O ₃	O ₂	O ₁	O ₀
0	0	0	0	0	0	0	0	0	0	I
0	0	1	0	0	0	0	0	0	I	0
0	1	0	0	0	0	0	0	I	0	0
0	1	1	0	0	0	0	I	0	0	0
1	0	0	0	0	0	I	0	0	0	0
1	0	1	0	0	I	0	0	0	0	0
1	1	0	0	I	0	0	0	0	0	0
1	1	1	I	0	0	0	0	0	0	0

Note: I is the data input



9-8 Demultiplexers (Data Distributors)

Security monitoring system using the 74ALS138.



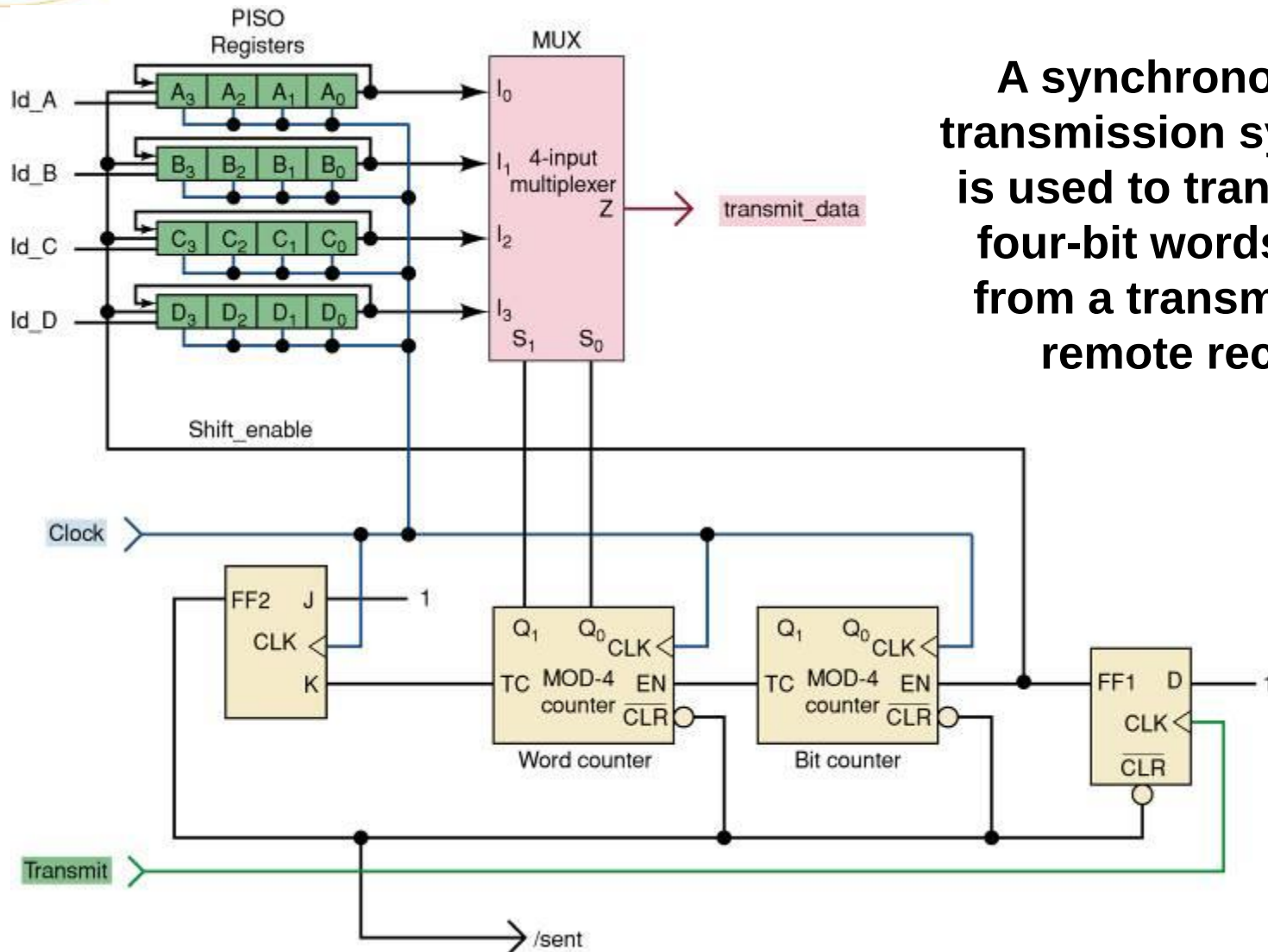
The system shown can handle eight doors, but can be expanded to any number.

The door switches are data inputs to the MUX.

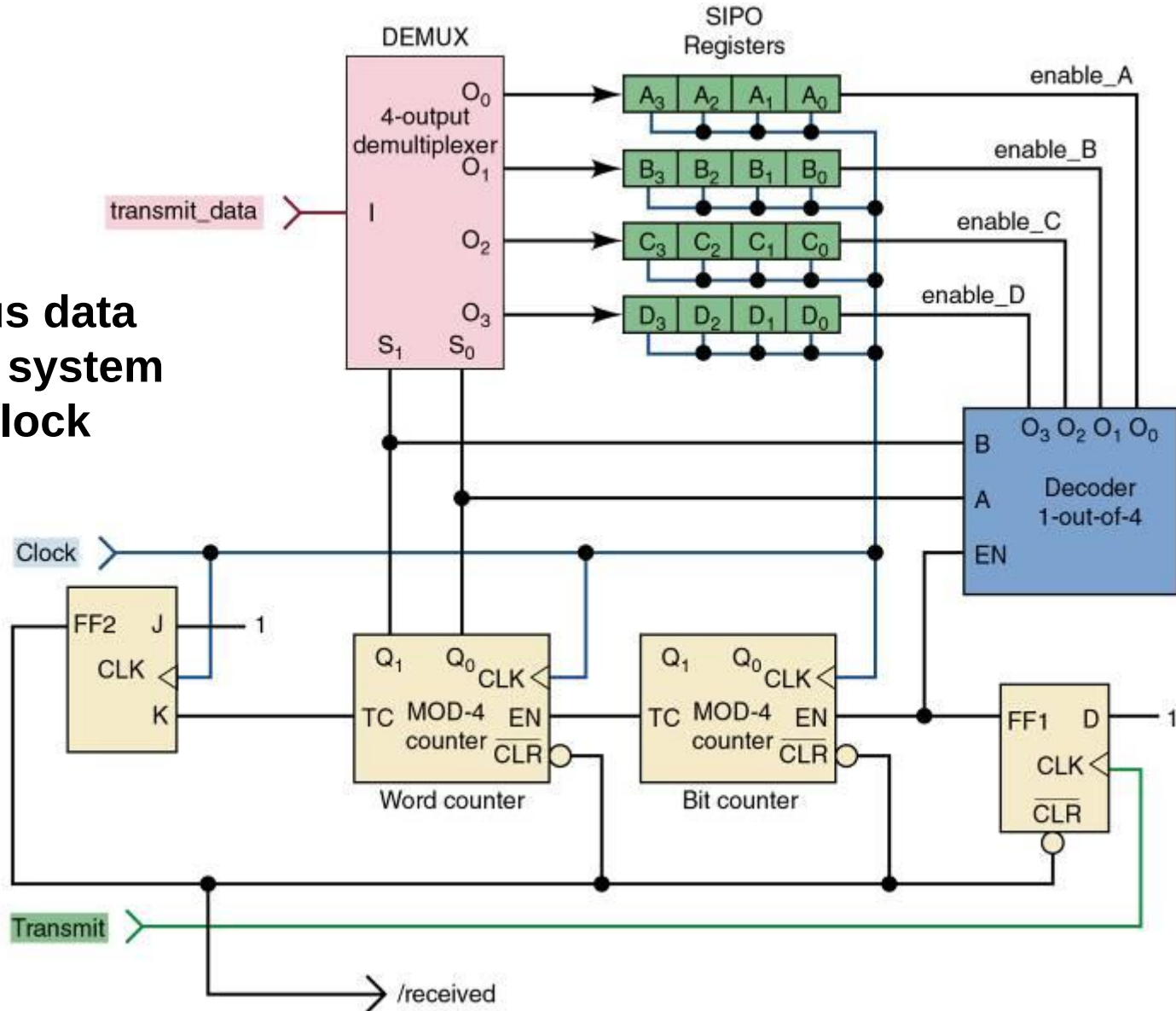
They produce a HIGH when a door is open and a LOW when it is closed.

9-8 Demultiplexers (Data Distributors)

A synchronous data transmission system that is used to transmit four, four-bit words serially from a transmitter to a remote receiver.

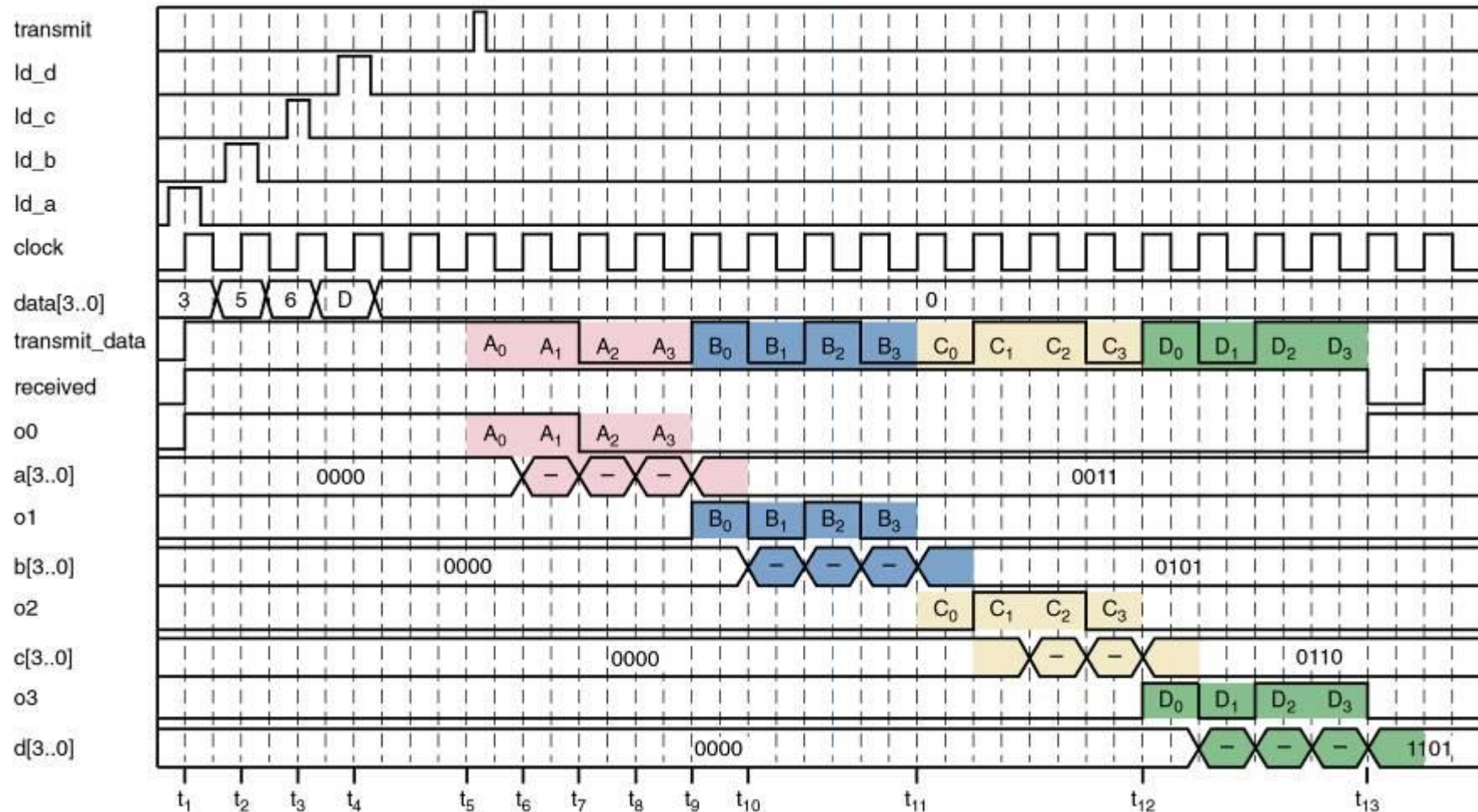


Synchronous data transmission system receiver block



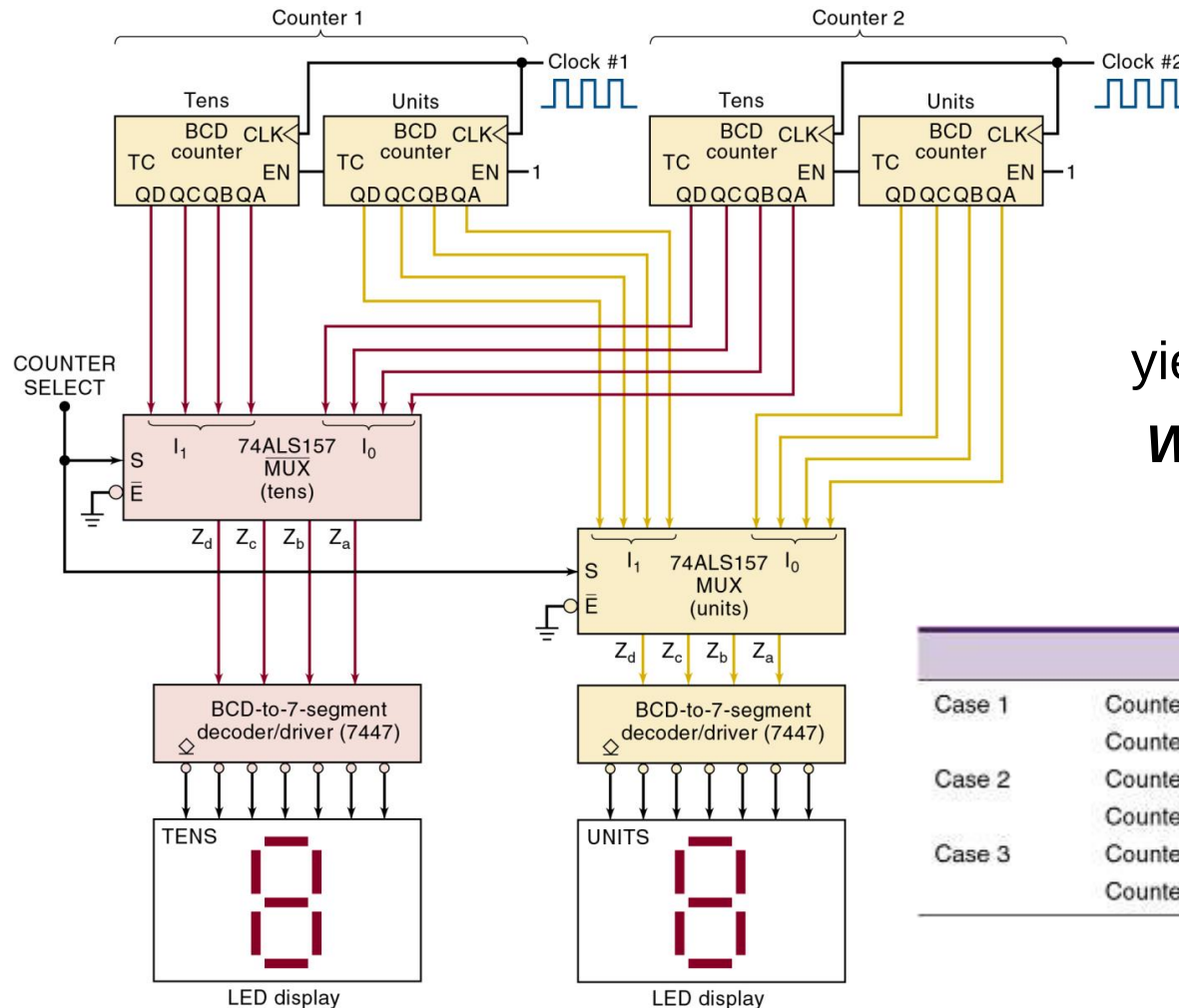
9-8 Demultiplexers (Data Distributors)

This timing diagram shows the parallel data loaded into the transmitter, the serial data stream, and distribution/storage of the four data values in the receiver registers.



9-9 More Troubleshooting

Apply observation and analysis to this example:



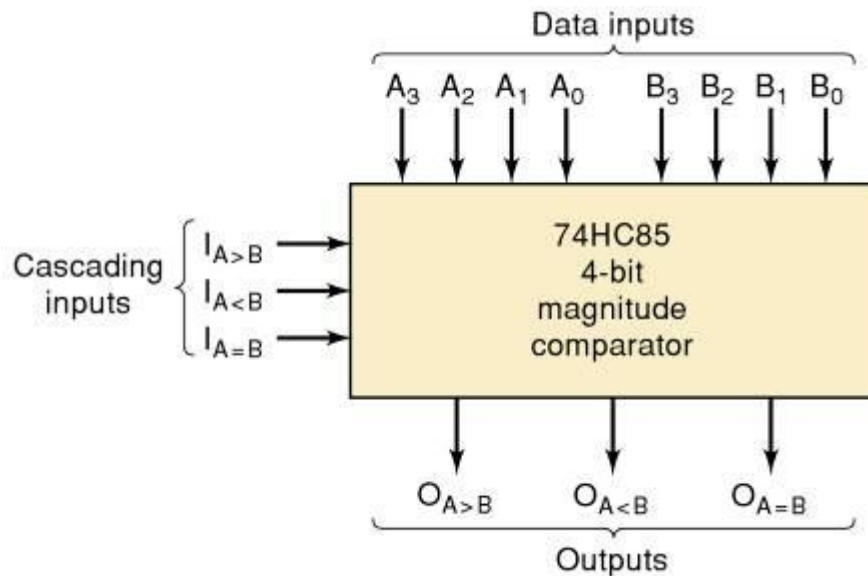
A test on the circuit yields the result shown.

What is the probable circuit fault?

		Actual Count	Displayed Count
Case 1	Counter 1	25	25
	Counter 2	37	35
Case 2	Counter 1	49	49
	Counter 2	72	79
Case 3	Counter 1	96	96
	Counter 2	14	16

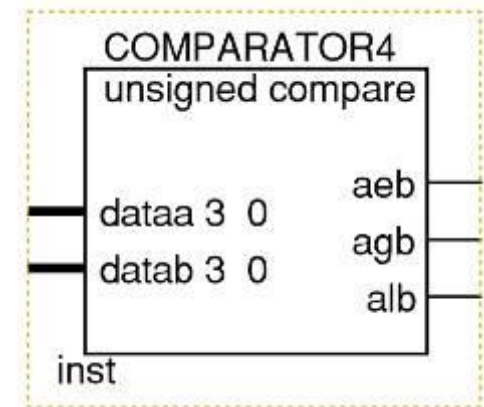
9-10 Magnitude Comparator

- Another useful MSI is a **magnitude comparator**.
 - A combinational logic circuit that compares two input binary quantities and generates outputs to indicate which one has the greater magnitude.



Cascading inputs are not necessary on a megafunction—simply specify larger data input ports.

Megafunction symbol.



9-10 Magnitude Comparator

- Another useful MSI is a **magnitude comparator**.
 - A combinational logic circuit that compares two input binary quantities and generates outputs to indicate which one has the greater magnitude.

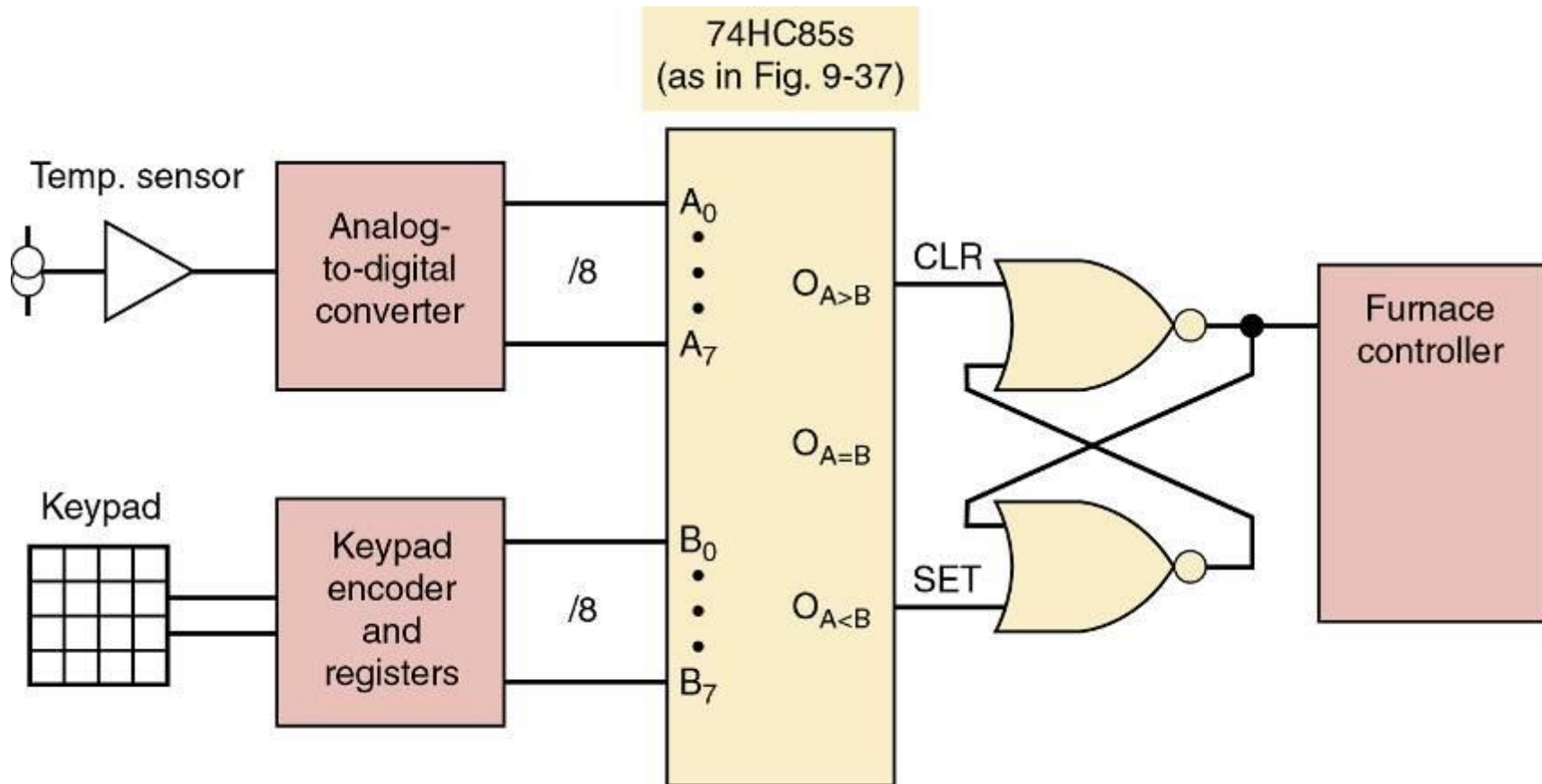
TRUTH TABLE

H = HIGH Voltage Level
L = LOW Voltage Level
X = Immaterial

COMPARING INPUTS				CASCADING INPUTS			OUTPUTS		
A ₃ , B ₃	A ₂ , B ₂	A ₁ , B ₁	A ₀ , B ₀	I _{A>B}	I _{A<B}	I _{A=B}	O _{A>B}	O _{A<B}	O _{A=B}
A ₃ >B ₃	X	X	X	X	X	X	H	L	L
A ₃ <B ₃	X	X	X	X	X	X	L	H	L
A ₃ =B ₃	A ₂ >B ₂	X	X	X	X	X	H	L	L
A ₃ =B ₃	A ₂ <B ₂	X	X	X	X	X	L	H	L
A ₃ =B ₃	A ₂ =B ₂	A ₁ >B ₁	X	X	X	X	H	L	L
A ₃ =B ₃	A ₂ =B ₂	A ₁ <B ₁	X	X	X	X	L	H	L
A ₃ =B ₃	A ₂ =B ₂	A ₁ =B ₁	A ₀ >B ₀	X	X	X	H	L	L
A ₃ =B ₃	A ₂ =B ₂	A ₁ =B ₁	A ₀ <B ₀	X	X	X	L	H	L
A ₃ =B ₃	A ₂ =B ₂	A ₁ =B ₁	A ₀ =B ₀	H	L	L	H	L	L
A ₃ =B ₃	A ₂ =B ₂	A ₁ =B ₁	A ₀ =B ₀	L	H	L	L	H	L
A ₃ =B ₃	A ₂ =B ₂	A ₁ =B ₁	A ₀ =B ₀	X	X	H	L	L	H
A ₃ =B ₃	A ₂ =B ₂	A ₁ =B ₁	A ₀ =B ₀	L	L	L	H	H	L
A ₃ =B ₃	A ₂ =B ₂	A ₁ =B ₁	A ₀ =B ₀	H	H	L	L	L	L

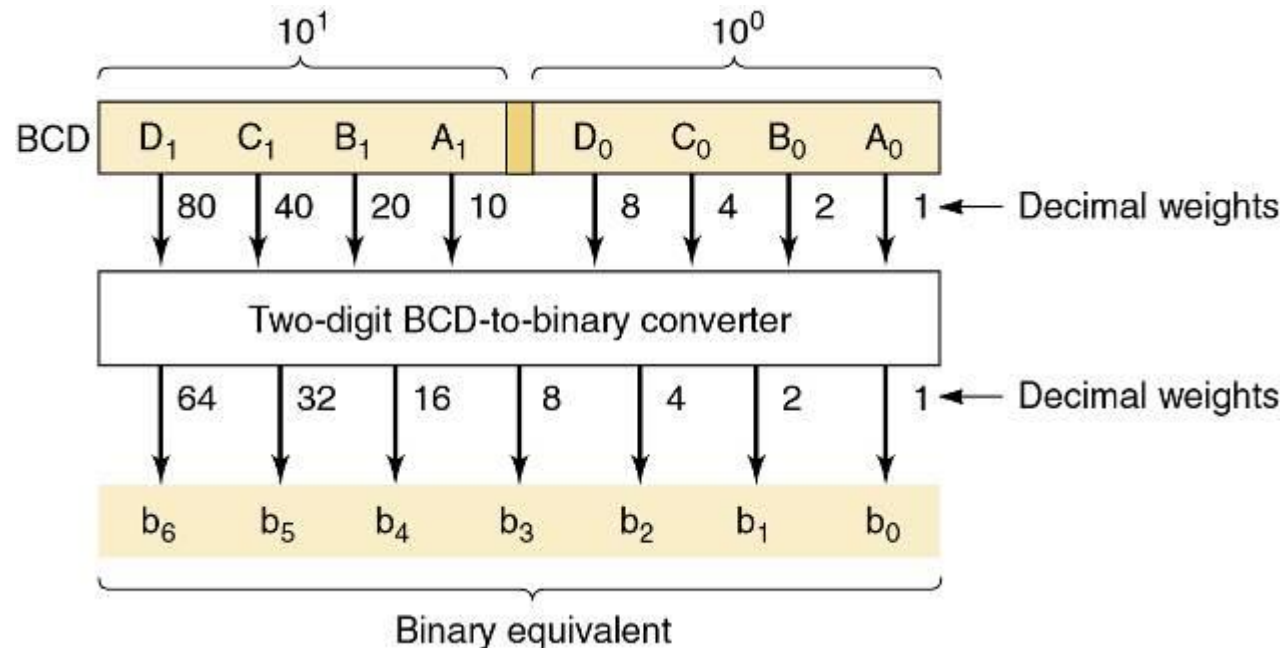
9-10 Magnitude Comparator

Magnitude comparator used in a digital thermostat.



9-11 Code Converters

- A code converter is a logic circuit that changes data presented in one type of binary code to another type of binary code.



Basic idea
of a two-digit
BCD-to-binary
converter.

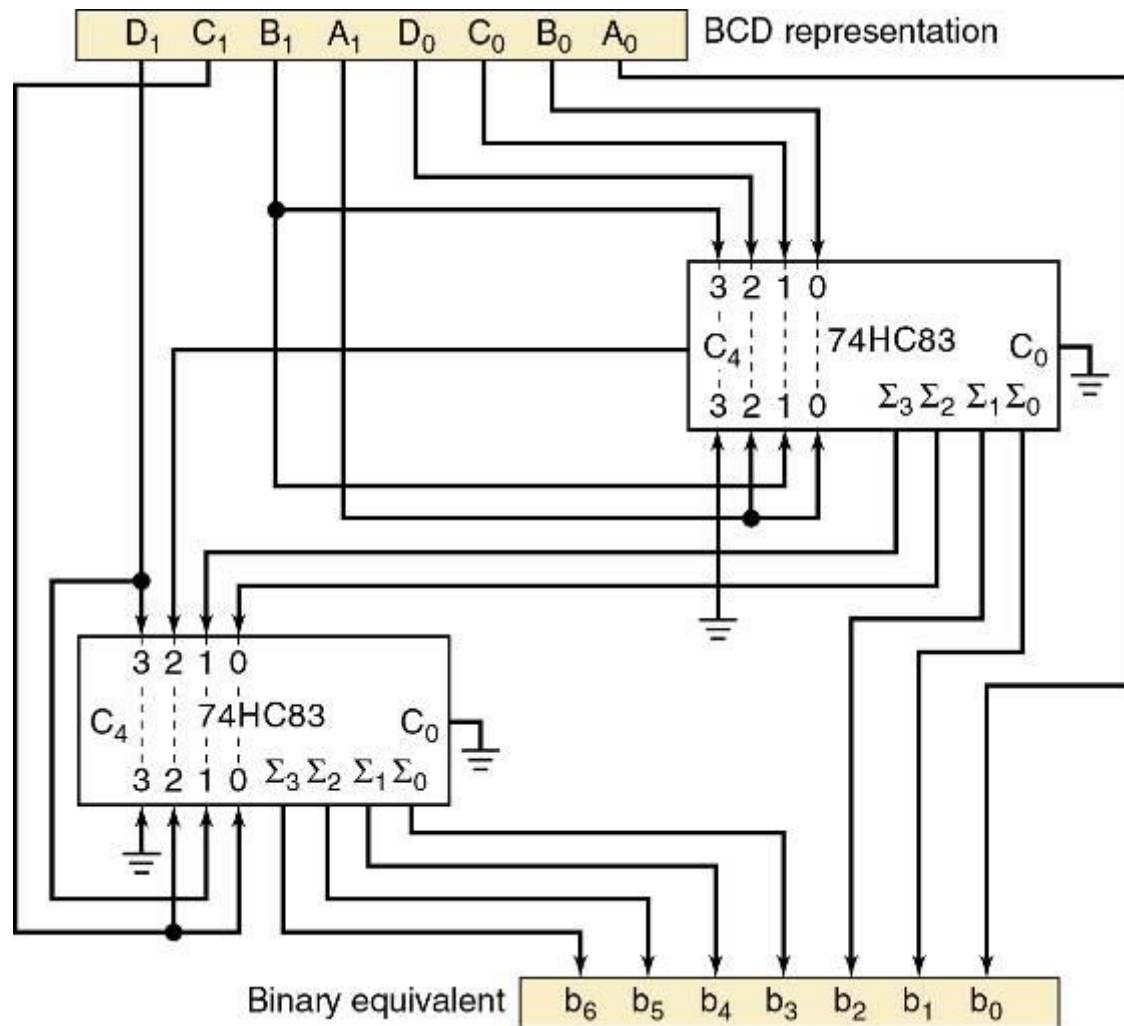
9-11 Code Converters

- The bits in a BCD representation have decimal weights that are 8, 4, 2, 1 within each code group.
 - That differ by a factor of 10 from one code group (decimal digit) to the next.

The decimal weight of each bit in the BCD representation can be converted to its binary equivalent.

BCD Bit	Decimal Weight	Binary Equivalent						
		b_6	b_5	b_4	b_3	b_2	b_1	b_0
A_0	1	0	0	0	0	0	0	1
B_0	2	0	0	0	0	0	1	0
C_0	4	0	0	0	0	1	0	0
D_0	8	0	0	0	1	0	0	0
A_1	10	0	0	0	1	0	1	0
B_1	20	0	0	1	0	1	0	0
C_1	40	0	1	0	1	0	0	0
D_1	80	1	0	1	0	0	0	0

9-11 Code Converters

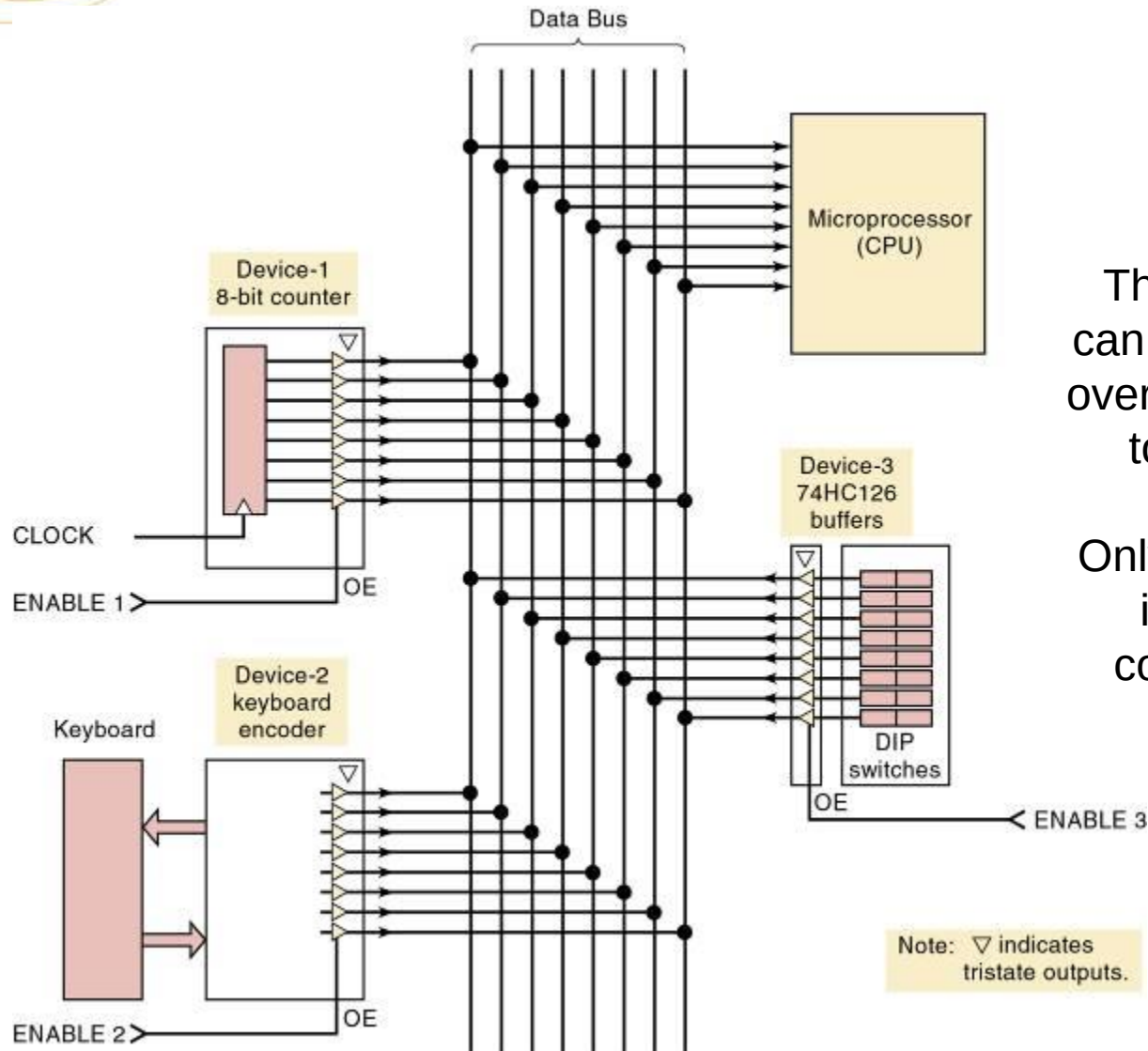


One way to implement the logic circuit that performs conversion process is to use binary adder circuits.

9-12 Data Busing

- In computers, transfer of data takes place over a common set of connecting lines called a **data bus**.
 - Devices tied to the data bus will often have tri-state outputs, or be tied to the data bus by tristate buffers.
- Devices commonly connected to a data bus:
 - Microprocessors; Semiconductor memory chips.
 - Digital-to-analog and analog-to-digital converters.

9-12 Data Busing



Three different devices can transmit eight-bit data over an eight-line data bus to a microprocessor.

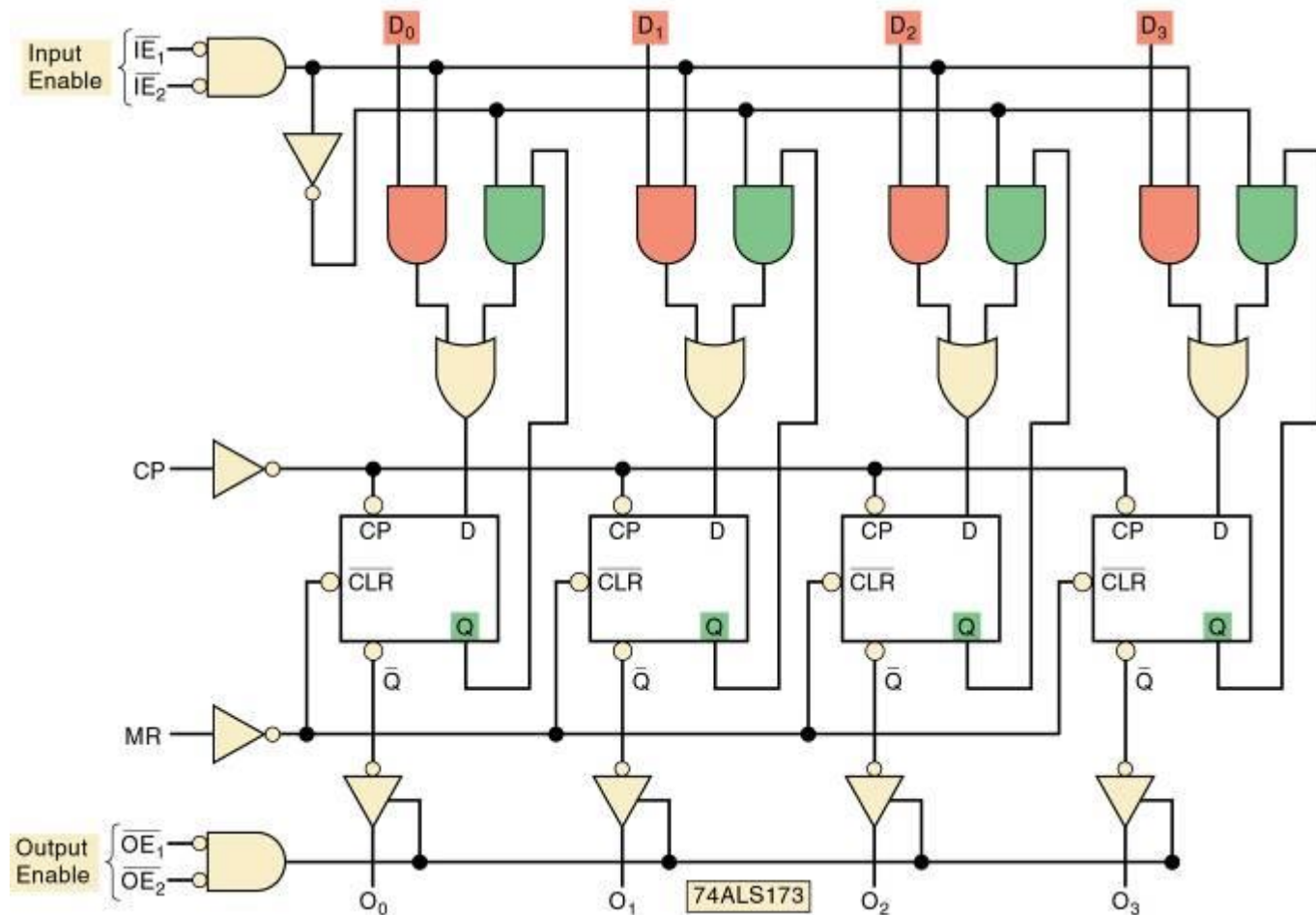
Only one device at a time is enabled—so bus contention is avoided.

9-13 The 74ALS173/HC173 Tristate Register

- The devices connected to a data bus will contain registers (usually flip-flops) that hold device data.
 - Outputs of these registers are usually connected to tristate buffers allowing them to be tied to a data bus.

9-13 The 74ALS173/HC173 Tristate Register

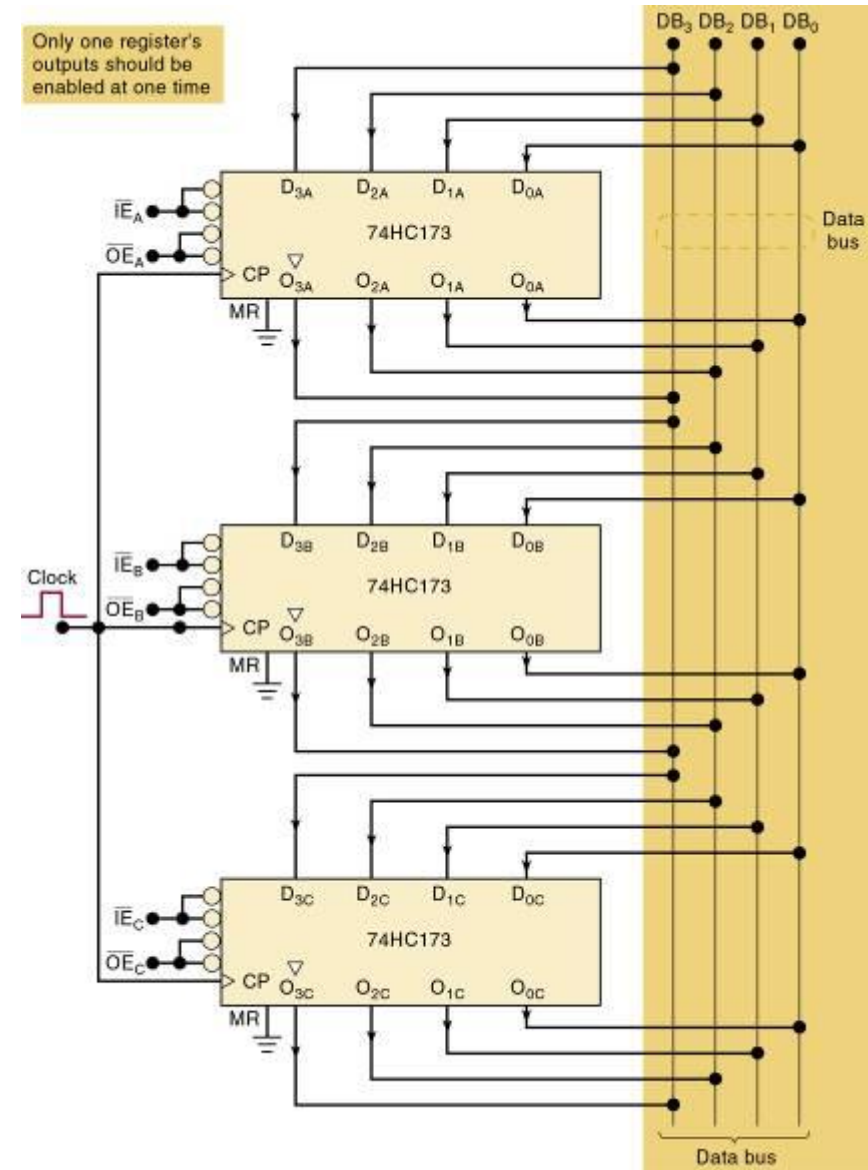
TTL 74ALS173 tristate register.



9-14 Data Bus Operation

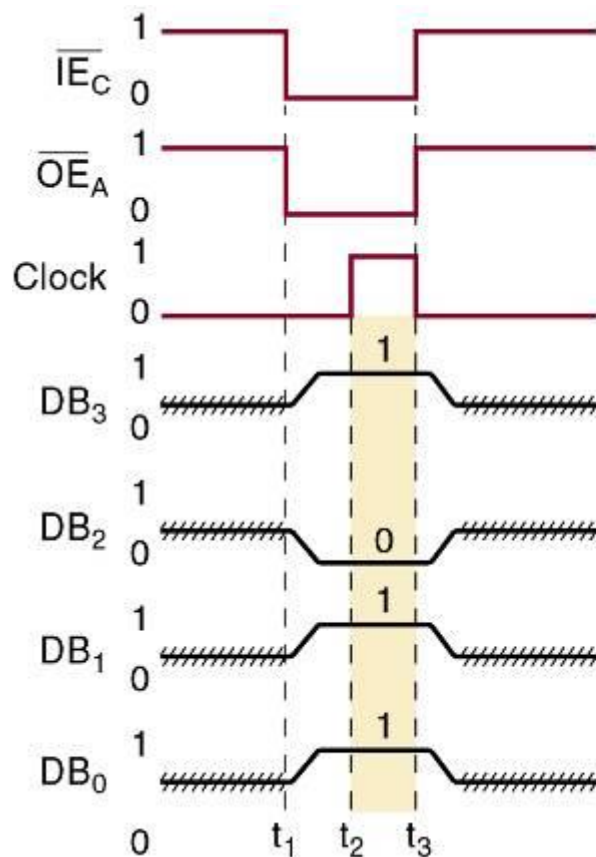
**Tristate registers
connected to a data bus.**

**The contents of any one of the
three registers can be parallel-
transferred over the data bus
to one of the other registers
through proper application
of logic levels to the
register enable inputs.**



9-14 Data Bus Operation

- The timing diagram shows various signals involved in the transfer of the data **1011** from register A to register C.



NOTES:

////// = floating (Hi-Z)

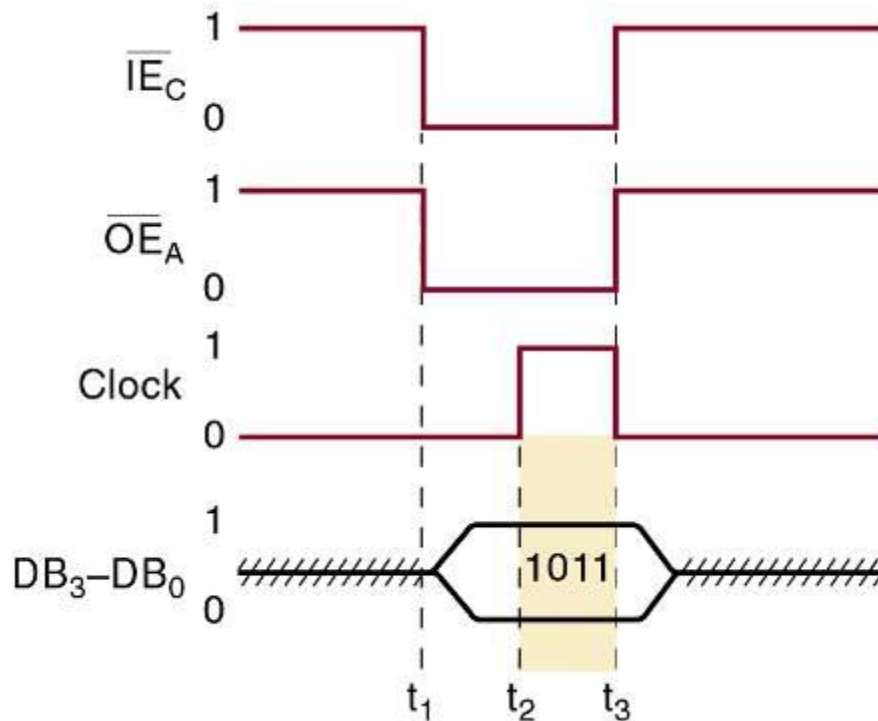
t_1 : Register A outputs are enabled. Its data are placed on the data bus lines.

t_2 : The PGT of the clock transfers valid data from the data bus into register C.

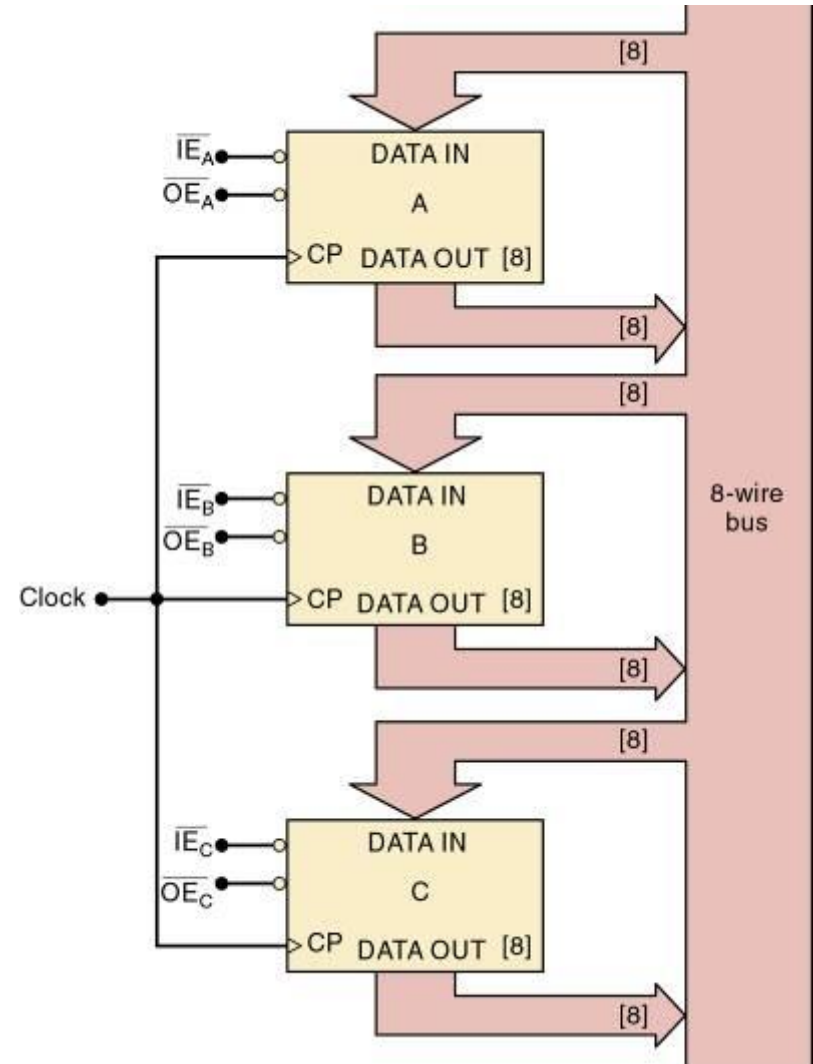
t_3 : Register A outputs are disabled and the data bus lines return to Hi-Z state.

9-14 Data Bus Operation

Simplified representation of bus arrangement.



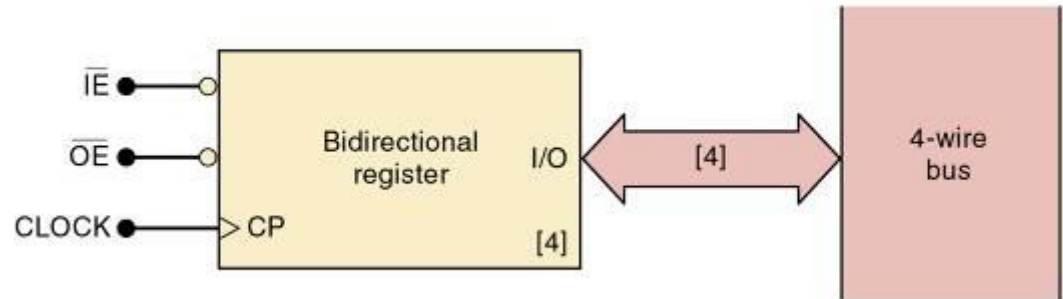
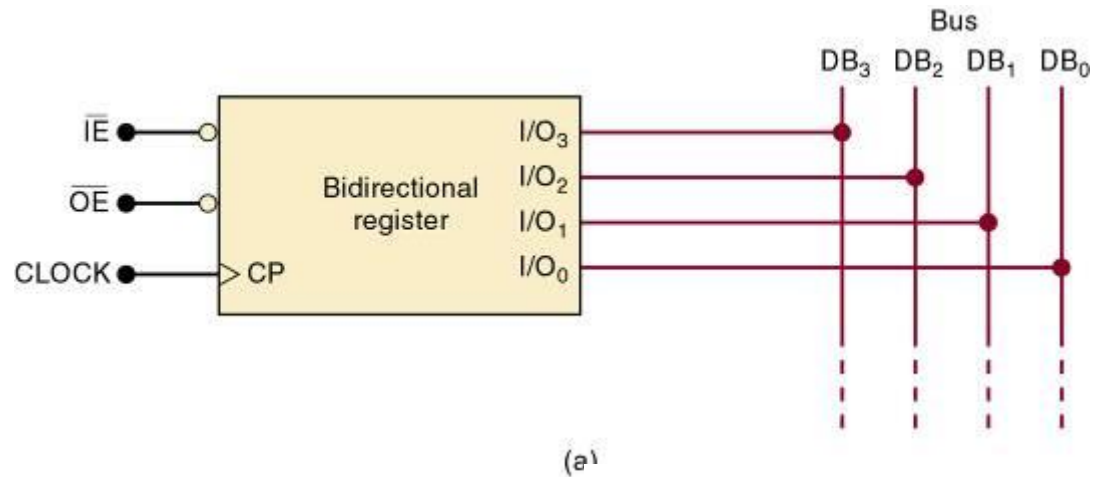
Simplified way to show signal activity on data bus lines.



9-14 Data Bus Operation

- Manufacturers have developed ICs that connect inputs & outputs *internal* to the chip.
 - To reduce the number of IC pins & bus connections.

Each I/O line will function as an input or an output depending on the states of the enable inputs—**bidirectional data lines.**



END

ELEVENTH EDITION

Digital Systems

Principles and Applications

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PEARSON